Hidden Violations or Costly Self-Reporting: Drinking Water Utilities' Mandatory Self-Reporting and Water Pollution

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Abstract

One model of regulation monitors adherence to the law by requiring the regulated to self-report their compliance. The regulated have an incentive to reduce how much they self-report in order to hide a violation of the law or avoid the costs of self-reporting. To target enforcement, regulators may require those who are more likely to violate the law to self-report more frequently. Under this regulatory model, the most likely violators have higher self-reporting costs and greater disincentives to self-report, perversely reducing information about the most likely lawbreakers. I document this perverse outcome among U.S. drinking water utilities that are exposed to water pollution from coal mining. Water pollution from coal mining creates variation in a utility's self-reporting costs and the likelihood of violating water pollution laws. I map utilities to coal mines using a hydrological network and create a dataset of utility self-reporting, water pollution, regulator inspections, and regulator enforcement. I use an instrumental variable to measure the effect of water pollution from coal production on utilities. Coal mining causes utilities to reduce self-reporting to cut costs rather than conceal water pollution violations. Coal mining increases regulator inspections at utilities, suggesting that regulators replace a portion of the monitoring that utilities do not undertake.

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Keywords: self-reporting; monitoring and enforcement; environmental regulation; drinking water quality; Safe Drinking Water Act; coal mining; water pollution; Acid Rain Program.

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1 Introduction

Many regulations in the United States rely on regulated entities to monitor themselves and submit mandatory self-reports of their compliance. This makes regulation more affordable to administer because it shifts the cost of gathering information from a resource-constrained regulator onto the regulated entities, which could number in the hundreds of millions of taxpayers self-reporting tax returns or tens of thousands of manufacturing firms self-reporting their handling of hazardous waste. The regulated entity has an incentive to save self-reporting costs by conducting less self-reporting, which I refer to as under-reporting. The regulated entity also has an incentive to under-report to conceal a violation of the regulation, especially if violations receive a penalty. To incentivize self-reporting, penalties are often issued for under-reporting. The decision to self-report depends, in part, on comparing the penalties for under-reporting with the cost savings from under-reporting and the avoided penalties from a self-reported violation.

Environmental pollution regulations in the United States often rely on regulated entities to monitor their own pollution levels and self-report their compliance to the regulator. When pollution is increasing, a regulated entity is more likely to violate a pollution standard and has an incentive to conceal that violation by under-reporting. When pollution is increasing, the regulated must increase the amount of self-reporting they conduct to provide regulators with information to protect public health or the environment. This increases the incentive to under-report to save costs. This regulatory design may perversely reduce the most pertinent monitoring when pollution increases. Whether self-reporting costs and concealing violations reduce self-reporting is an empirical question that has implications for designing any self-reporting regulation.

This paper studies the impact of self-reporting costs and the incentive to conceal a legal violation on a firm's propensity to self-report its compliance with a regulation. In addition, I study how increasing the amount of required self-reporting to gain more information affects self-reporting. In particular, I study U.S. drinking water utilities' mandatory water pollution self-reporting between 1985 and 2005. I identify changes in self-reporting costs and the likelihood of violating water pollution standards using changes in a utility's exposure to water pollution from coal mining. When water pollution increases, utilities are required to conduct more self-reporting.[1] This setting is appropriate to study mandatory self-reporting since the Safe Drinking Water Act ("SDWA") requires utilities to treat and self-report their water pollution to the regulator. Utilities that violate

1. In a similar vein to the increased self-reporting requirements of drinking water utilities, the Clean Air Act exempts very small emitters, who pollute less than larger emitters, from reporting requirements.

a water pollution regulation receive costly enforcement action or are required to make costly investments in improving water treatment. Utilities find self-reporting is costly, but under-reporting results in a penalty. Coal mining creates variation in the costs of self-reporting and the likelihood of violating a regulation while being a source of variation that is separate from the drinking water utilities. Coal mines and drinking water utilities are managed separately, so utilities cannot change water pollution from coal mining.

I map U.S. coal mines to drinking water utilities using a hydrological flow map and build a dataset of individual coal mine production, coal bed geochemistry, utility pollution levels, utility self-reporting, utility regulation violations, regulator enforcement, and regulator visits. I first establish that coal mining pollutes water at drinking water utilities. I find that coal mining increases the levels of inorganic chemical ("IOC") pollution from arsenic and nitrates at drinking water utilities. I use placebo tests and extensive fixed effects to find evidence that coal mining affects utilities by changing water pollution and not through another channel.

I use an instrumental variable to measure the impact of changes in coal mining on utility self-reporting. I find that coal mining reduces utility self-reporting. I find that utilities reduce self-reporting to avoid self-reporting costs rather than to conceal violations of water quality standards. I conclude that utilities are not concealing violations since I do not observe pollution readings that violate water quality regulations. In addition, I find that a utility is more likely to under-report when self-reporting costs increase and the utility's pollution levels are within legal water quality standards. Finally, regulators increase inspections at utilities with more pollution and under-reporting, but enforcement does not change at those utilities. Utilities would receive significant penalties if they were discovered to have concealed violations. This suggests that utilities are not systematically under-reporting to conceal water quality violations.

I use an instrumental variable for changes in coal production since I do not have sufficient data to measure all utility characteristics that affect the number of coal mines a utility is exposed to and a utility's self-reporting. A poorly managed utility will under-report more often and be less successful in lobbying regulators to prevent mines from opening in their water sources. This results in less well-managed utilities being exposed to more coal mines and under-reporting more. Without data on utility management, I would overestimate the impact of coal mining on utility under-reporting.

I adapt an instrument for coal mining from Douglas and Wiggins (2015) and apply it to drinking water utilities' exposure to coal mining for the first time. The instrumental variable measures changes in coal production that resulted from national trends in coal demand and were unrelated to a utility's characteristics. Coal-fired power plants are the

main consumers of coal and have preferences over the chemical properties of the coal they use. After the federal government enacted the Acid Rain Program ("ARP") in 1990 to limit sulfur emissions from coal-fired power plants, the demand for coal containing a lower amount of sulfur increased. The ARP and the timing of its enactment were unrelated to drinking water quality or drinking water utility management. The effect of the ARP on coal mining and its timing is plausibly independent of utility characteristics associated with self-reporting water quality; therefore, the instrument estimates the causal effect of coal production on drinking water utility behavior.

The causes of under-reporting are a first-order question for regulatory design since compliance monitoring is the first step in enforcing regulation. Under-reporting undermines several effective enforcement strategies that utilize violation history, which the enforcement literature has identified. The literature finds that regulators improve compliance with environmental regulations by issuing larger sanctions for larger violations (Shimshack 2014; Harford 1987), escalating sanctions for repeat offenders (Blundell, Gowrisankaran, and Langer 2020; Blundell 2020), and selectively targeting the worst polluters (Duflo et al. 2018). These approaches to enforcement require monitoring information, which the regulator lacks when self-reporting firms under-report. I show that regulation which increases the costs of self-reporting can lead to increased under-reporting and undermine these enforcement strategies.

Self-reporting is costly, and since the monitoring requirements under the SDWA increase as water quality deteriorates, the costs utilities face to comply with self-reporting also increase with pollution. Across this literature, the firm's self-reporting cost is not modeled, nor is the way self-reporting costs change with pollution. I relax these assumptions in this paper and model firms' decisions to self-report when self-reporting is costly and under-reporting can hide a violation. The cost-of-reporting channel has been identified as a deterrent to self-reporting in tax compliance (Tazhitdinova 2018). This is the first paper to identify the cost-of-reporting channel as a cause of under-reporting in environmental regulation monitoring. The literature has primarily modeled under-reporting as firms strategically withholding information to hide a violation. In work on water utilities, under-reporting is attributed to hiding a water quality violation (Naylor 2024; Andarge et al. 2025; Wallsten and Kosec 2008), with Bennear, Jessoe, and Olmstead (2009) being a rare example of over-reporting to hide a water quality violation. Lee (2020) finds that oil and gas firms under-report to hide a regulatory violation. A closely related literature finds that regulated entities exploit gaps in monitoring coverage rather than misstating measurements. Ambient air quality deteriorates on days when monitors are not scheduled to operate (Zou 2021); firms shut down pollution monitors strategically during times of

increased output (Mu, Rubin, and Zou 2026); and small cement plants can remain exempt from air quality reporting, go unmonitored, and cause large health damages (Zirogiannis et al. 2023). Work on how to induce self-reporting asks what combination of audit frequency and penalties makes firms self-report a violation despite the incentive to hide a violation (Innes 1999; Foster and Gutierrez 2013; Anton, Deltas, and Khanna 2004), with Malik (1993) finding that infrequent visits paired with high penalties induce better self-reporting. I show evidence that inspections without penalties are insufficient to induce self-reporting.

Designing regulations to induce self-reporting is important for any self-reported regulation, but it is particularly important for producing reliable measurements of U.S. water quality. According to self-reported drinking water quality data, between 9 and 34 million Americans each year are served by a drinking water utility that violated a health-based drinking water standard (Allaire, Wu, and Lall 2018). U.S. drinking water quality has improved substantially since the SDWA (Keiser et al. 2026), with notable exceptions such as the crisis in Flint, Michigan (Christensen, Keiser, and Lade 2023), and significant remaining health costs from polluted drinking water (Hill and Ma 2017; Keiser and Shapiro 2019; DiSalvo and Hill 2024; Marcus 2025). However, much of this evidence comes from utility self-reports and is therefore unable to measure water quality that was never reported. A related literature argues that U.S. water quality regulation has improved to the point where the remaining gains from abating point-source pollution are small, since eliminating point-source emissions entirely would improve only 10 percent of U.S. rivers and streams in EPA quality ladder rankings (Bingham et al. 2000). However, the benefits of surface water regulation may be undervalued due to the difficulty of measuring water pollution — a problem that instrumental variables can address (Keiser 2019). My findings suggest that even small amounts of point-source pollution can lead to negative impacts on drinking water quality.

The environmental science literature has linked coal mining to ambient water pollution (Skousen, Ziemkiewicz, and McDonald 2019; Akcil and Koldas 2006), and the economic literature has documented coal mining's effects on air quality, greenhouse gas emissions, and health (Muller, Mendelsohn, and Nordhaus 2011; Parfitt 2024); however, this work has not established whether coal mining pollution reaches treated water at drinking water utilities and, through them, households. The findings have contemporary relevance since U.S. federal energy policy has expanded mining access on federal lands and intervened to prevent planned coal-fired power plant retirements (U.S. President (Trump) 2025a, 2025b, 2025c; Daly 2025).

This paper makes three contributions. This is the first paper to identify self-reporting

costs rather than violation concealment as a reason a firm does not self-report when self-reporting is mandatory. Self-reported compliance underpins regulations as diverse as income taxes and public company finances, and wherever the cost of self-reporting rises, monitoring weakens. This is especially important when self-reporting costs rise for those most likely to break the law; this leads to a reduction in monitoring information when the information is most valuable to regulators. This is the first study to identify that the cost of self-reporting, rather than the concealment of a water quality violation, is a cause for utilities to under-report water quality. Finally, I provide the first causal estimates of coal mining's effect on drinking water quality at treated utilities in the United States, identifying that coal mining pollutes piped household water.

The rest of the paper proceeds as follows. Section 2 sets out the model of self-reporting. Section 3 describes the SDWA, the self-reporting and enforcement institutions it creates, coal mining and water quality, and coal mining and the Acid Rain Program. Section 4 describes the data. Section 5 estimates the effect of coal mining on water pollution at utilities. Section 6 constructs the instrument for coal mining and assesses its validity. Sections 7 and 8 report the effects of coal mining on utility self-reporting and water quality violations, as well as on regulator visits and enforcement. Section 9 discusses the mechanisms, and Section 10 concludes.

2 Theoretical model

2.1 Setup

I build a theoretical model to formalize a utility's decisions to comply with water pollution regulation while experiencing pollution from an external source. Utilities are required to meet pollution requirements and self-report compliance. I follow Mookherjee and Png (1994) and assume that utilities find pollution compliance costly. Compliance cost is determined by a utility's ability, such as the skills of its managerial and technical staff, and the costs of water treatment, including the cost of filtration equipment and sterilization chemicals. A utility's compliance cost, θ, is a random variable drawn from $F(\cdot\,; m)$, a strictly increasing and continuously differentiable cumulative distribution function with support $(0, \bar{\theta})$ and density $f(\cdot\,; m)$, which depend on pollution m. A larger θ represents higher compliance costs. Increasing m weakly lowers the probability that a utility's compliance cost falls at or below any given level, so that $\partial F(\theta; m)/\partial m \leq 0$. Utilities know their draw of θ, but regulators only know the distribution of θ.

Utilities receive a benefit from neglecting their pollution control requirements. The

utility chooses a negligence level $a \in [0, \bar{a}]$, which measures how much it avoids the requirements of drinking water quality regulations, ranging from full compliance, $a = 0$, to avoiding every requirement, $a = \bar{a}$. Regulators do not directly observe a utility's choice of a. A utility receives a benefit $\theta b(a)$ from having compliance costs θ and choosing a negligence level a, where $b'(a) > 0$ and $b''(a) < 0$. A utility's benefits from negligent pollution control are the savings on managers' time, wages, and chemicals. The opportunity cost to the utility of choosing a negligence level a relative to maximum negligence $\bar{a}$ is $\theta[b(\bar{a}) - b(a)]$. The financial benefit of a certain level of negligence to a profit-maximizing, privately owned utility is obvious. Publicly owned utilities also benefit from cost savings that allow them to forgo rate increases or tax collection, which may be unpopular with residents.

Utilities are legally required to self-report their water quality each year but have the choice to under-report. Given negligence a, a year of self-reporting would reveal a water quality violation with probability $q(a)$, with $q(0) = 0$, $q' > 0$, and $q'' \geq 0$. I modify the penalty structure from Kaplow and Shavell (1994) by assuming that self-reporting is costly. A utility's annual self-reporting cost $c \sim G(\cdot\,; m)$ is drawn from a strictly increasing and continuously differentiable cumulative distribution function with density $g(\cdot\,; m)$ and support $[0, \bar{c}]$, where the support itself does not vary with m. Regulators require utilities to increase sampling as pollution increases, so the distribution $G(\cdot\,; m)$ depends on upstream pollution m with $\partial G(c; m)/\partial m \leq 0$ for every c. Raising m weakly lowers the probability that a utility's self-reporting cost is at or below any given level.

Regulators perfectly observe under-reporting. Unlike Mookherjee and Png (1994), I assume regulators do not observe water quality violations unless utilities self-report water quality or the regulator conducts an inspection. Regulators use penalties and investigations for under-reporting and water quality violations. Utilities receive a sanction r if they self-report a pollution level above the legal pollution limit and a penalty t for under-reporting. Regulators investigate under-reporting utilities with a probability p and issue a sanction s for a discovered water quality violation, where $s > r$.

I maintain two assumptions on the sanctions throughout. First, $r > ps$: the certain sanction a utility bears for disclosing a water quality violation exceeds the expected sanction it bears for concealing one, which gives the utility an incentive to conceal. Together with $s > r$, this requires the investigation probability to satisfy $p < r/s$. Second, $t \geq (r - ps)\, q(\bar{a})$, which keeps the self-reporting threshold defined as non-negative at every negligence level. Without it, $(r - ps)q(a)$ could exceed t at high negligence, and since self-reporting costs satisfy $c \geq 0$, a negative threshold would mean no utility with any cost draw self-reports.

I maintain one regularity condition on the negligence choice: b and q are such that optimal negligence is interior, $a \in (0, \bar{a})$, at every type in the support of θ and at either marginal sanction, r or ps. Interiority makes the first-order condition below both necessary and sufficient within a decision to self-report or under-report.

Each year the utility chooses either to self-report at a cost c or to not report and receive an under-reporting violation. Expected costs are the sum of expected penalties and self-reporting costs:

$$\begin{aligned} \text{self-report:} \quad & c + q(a)\, r \\ \text{under-report:} \quad & t + p\,[q(a)\, s] \end{aligned}$$

The utility self-reports if and only if

$$c \leq \hat{c}(a) \equiv t - (r - ps)\, q(a). \tag{1}$$

The term $(r - ps)q(a)$ is the wedge between self-reporting and not reporting a water quality violation. Even a system with no chance of a water quality violation, $q(a) = 0$, stops self-reporting once $c > t$, that is, when self-reporting is more expensive than the sanction for skipping it, even though there is no water quality violation to conceal. The self-reporting threshold is decreasing in $q(a)$, so systems with a higher probability of violating a water quality rule reduce self-reporting at lower cost thresholds. The expected annual cost of the reporting decision, at a cost draw c, is

$$e(a; c) = \min \left\{ c + q(a)r, \;\; t + \, q(a)ps \right\}, \tag{2}$$

which is increasing in a.

2.2 Model predictions

The model predicts that the share of utilities under-reporting rises with pollution, and it does so through two channels, neither of which requires a utility to be hiding a water quality violation.

The first is a pure cost channel. A utility self-reports when its draw of the self-reporting cost falls below the cutoff $\hat{c}(a)$ in equation (1), which compares the cost of self-reporting against the penalty for under-reporting, net of what the utility saves by not revealing a water quality violation. Regulators require more frequent self-reporting from utilities whose source water is impaired, so pollution shifts the distribution of

self-reporting costs to the right while leaving each utility's cutoff unchanged. More of the population lands above its own cutoff, and under-reporting rises. This channel operates even for a utility whose water quality does not approach violation levels, since such a utility still stops reporting once self-reporting costs more than the penalty for not reporting, and it has no water quality violation to conceal.

The second channel works through the composition of compliance costs. Pollution also raises the cost of treating water to any given standard, shifting the distribution of the compliance-cost type θ to the right. Utilities with higher compliance costs choose more negligence, and a more negligent utility faces a higher probability that a self-report would reveal a water quality violation, which lowers its own reporting cutoff. Pollution therefore moves the population toward types that are already closer to not self-reporting. The comparative statics of both channels are non-negative, so the share of utilities under-reporting is non-decreasing in pollution. This is Proposition 2, and it is one of the predictions tested in the empirical work.

The model predicts that the probability of a water quality violation increases when pollution increases, conditional on a utility conducting a self-reported water quality test. By assumption, pollution raises compliance costs. Compliance costs raise negligence since the benefits of negligence are scaled by the compliance cost. Finally, the probability of a water quality violation upon self-reporting increases with negligence.

Appendix A.1 states and proves the three propositions: negligence is increasing in compliance costs (Proposition 1), the share of under-reporting utilities is non-decreasing in pollution (Proposition 2), and scaling every sanction by a common factor raises self-reporting (Proposition 3). Proposition 3 is the model's policy statement and is not tested here.

3 Institutional background

Empirically identifying the impact of pollution on firm behavior requires separating the firm's choices over output and pollution. Separating a firm's output and pollution in most industries is challenging since firms can reduce pollution by reducing output. For example, in the U.S., the Toxics Release Inventory requires firms to report emissions, handling, and use of chemicals above a reportable quantity. Firms can avoid the reporting requirement by emitting less or producing less, and they respond strategically to such reporting thresholds (Bennear 2005). Water pollution from coal mining is an external process for drinking water utilities since they do not directly control coal production, making it an ideal setting to study the effect of pollution on self-reporting.

3.1 Drinking water utilities, regulators, and the Safe Drinking Water Act

I focus on drinking water utilities that are regulated by the SDWA and had water intakes downstream of coal mines between 1985 and 2005. These utilities serve at least 25 year-round residents or have at least 15 service connections, are required to remove water pollution for a list of regulated pollutants, and self-report water quality tests to state regulators. Each regulated pollutant has a maximum contaminant level ("MCL"), the maximum concentration of the pollutant permitted in drinking water; a utility's pollution level cannot legally exceed it. When this limit is exceeded, the utility receives an MCL violation. MCL violations are serious threats to public health, require timely rectification, and often trigger enforcement actions. Regulators primarily rely on utilities to self-report water quality, creating an opportunity for utilities to conceal an MCL violation by not reporting. Utilities that fail to report or under-report receive a monitoring and reporting ("MR") violation. Pollution increases a utility's likelihood of exceeding the MCL, increasing the incentive to under-report in order to conceal an MCL violation.

Utilities have an additional incentive to under-report to avoid self-reporting costs. Self-reporting is expensive in itself and requires hiring technical and administrative staff, as well as laboratories. A survey of Indiana drinking water utility managers in 1980 found that the reporting requirements of the SDWA had the greatest impact on their systems (Oleckno 1982). Utilities found self-reporting requirements difficult to meet due to the technical burden of properly collecting the samples, storing them, and then shipping them to a lab. They also found the administrative burden of conducting the correct tests on time to be burdensome. In 1992, utilities were required to test for 82 pollutants. The baseline schedule varied by pollutant and system type. For example, groundwater utilities were required to test for nitrates once a year, and utilities serving fewer than 1,000 people had to test for coliforms at least once a month.

Self-reporting requirements, and therefore self-reporting costs, increase with pollution. Regulators require utilities with polluted water supplies, or those at risk of pollution, to conduct more frequent tests. For example, utilities that detected nitrates above 50% of the MCL (5 mg/L during the period of this study) were required to increase testing from once a year to quarterly, a fourfold increase in nitrate testing work.

States and tribes conduct enforcement actions against utilities that violate MCLs or receive MR violations. The SDWA gives states wide discretionary powers to tailor compliance and enforcement plans to each utility and violation. The intention is for the violation to be resolved quickly enough to reduce the health impacts of pollution. MCL vi-

olations are treated seriously and receive stricter enforcement than MR violations. However, an intentional MR violation concealing an MCL violation is punished more harshly than a regular MR violation. Enforcement is frequently remedial rather than punitive, addressing the cause of the violation. Enforcement can be informal (letters, notices of violation, compliance plans) or formal (court-mandated consent decrees, legal prosecution, fines). Enforcement without financial penalties is still costly to utilities. Public notifications for violations have large reputational costs that deter violations (Bennear and Olmstead 2008). Consent decrees can require installing expensive treatment infrastructure or hiring technical personnel. The replacement of an existing arsenic filtration system at a utility with baseline arsenic levels of 0.029 mg/L, serving 250 people in Pennsylvania in 2011, cost $191,970, including $136,744 for equipment, $21,726 for site engineering, and $33,500 for installation. In addition, it cost an average of $60 a day in chemical, electricity, and labor costs (U.S. Environmental Protection Agency 2011).

Regulators also enforce utility compliance through on-site visits, which the EPA categorizes into five groups based on the reason for the visit: sanitary, technical assistance, enforcement, sample collection, and inspection visits (U.S. Environmental Protection Agency 2026b). The most extensively documented of these is the sanitary survey, an on-site evaluation conducted by the state or tribal agency, of eight components of a utility's system: its water source, treatment process, distribution system, finished-water storage, pumps and pumping controls, monitoring and reporting practices, system management and operation, and operator compliance with certification requirements (U.S. Environmental Protection Agency 2026a). Utilities receive a sanitary survey at least once every three years; the survey also gives the primacy agency an opportunity to educate operators on proper monitoring and sampling procedures and to provide technical assistance (U.S. Environmental Protection Agency 2026a). Technical assistance visits involve engineering plan review and operations-and-maintenance guidance, and sample collection visits involve the regulator collecting the water sample.

Enforcement visits accompany a formal enforcement action, an investigation of a complaint or violation, or an emergency response, while inspection visits comprise site inspections, regularly scheduled inspections, and informal inspections outside the sanitary-survey cycle (U.S. Environmental Protection Agency 2026b). A 2024 EPA fact sheet on utility inspections describes the general on-site inspection process: inspectors open with a conference, walk through the facility to examine the source, treatment, distribution, and storage components, and the condition of equipment, collect water samples, and review the utility's monitoring plans, sampling logs, operational manuals, and distribution system maps (U.S. Environmental Protection Agency 2024). Inspections may also follow

up on a prior inspection, sanitary survey, enforcement action, or citizen complaint; EPA typically issues an inspection report within 60 to 90 days, and further enforcement action may follow if violations are identified (U.S. Environmental Protection Agency 2024).

3.2 The Acid Rain Program (ARP) and utility exposure to coal mining

Arsenic, nitrates, barium, and selenium are common pollution byproducts of coal mining that enter water through acid mine drainage ("AMD") (Liang et al. 2017; Boente, Baragaño, and Gallego 2020; Rouhani, Skousen, and Tack 2023). AMD occurs when rocks containing sulfides are exposed to air and water, beginning the process of oxidation and decomposition (Ceto and Mahmud 2000). The oxidized minerals form sulfuric acid, which dissolves metal ions and sulfates that can then be carried into surface water or groundwater (Akcil and Koldas 2006; Skousen, Ziemkiewicz, and McDonald 2019). Mines can reduce AMD by covering excavated material with water in tailings ponds. Coal mining can also generate pollution through coal wash slurry. Coal washing processes coal with water and chemicals to remove impurities and improve combustibility. The slurry waste is often mixed with water and stored in retaining ponds or abandoned mine shafts. Dissolved pollutants enter groundwater and surface water by seeping through porous rocks or through retaining pond failures (Sealey 2000).

I focus on the SDWA regulation of arsenic, nitrates, barium, selenium, and general IOCs since they can be pollution byproducts of coal mining, and their MCLs did not change between 1985 and 2005.[2] Nitrate and arsenic self-reporting requirements changed nationally between 1985 and 2005. To address the concern that changes in MR violations would be the result of changing self-reporting requirements, I control for these changes with year fixed effects in empirical models. The SDWA IOC rule contains all regulated IOCs during this period, some of which are subject to changing MCLs and some of which may be pollution byproducts of coal mining.

During the sample period, coal fired power plants increased their demand for low sulfur coal, coal containing less than 1.5% of sulfur by weight, in order to meet new federal clean air regulations known as the ARP. The ARP was signed on November 15, 1990, as an amendment to the Clean Air Act, with the intention of reducing acid rain by capping emissions of sulfur dioxide (SO2) and nitrogen oxides (NOx), acid rain precursors, from coal-fired power plants. Power plants were fined if their emissions exceeded their allowances. The ARP achieved 100 percent compliance (Lattanzio 2022). Although com-

2. The list of regulated pollutants has been added to over time: in 1985, there were 23 regulated pollutants, and by 2005, there were 93.

pliance could be achieved through allowance trading, SO2 scrubbers, and NOx scrubbers, fuel switching to low-sulfur coal was the primary abatement response.

Emissions were capped in two phases. In Phase I, the EPA issued emission permits and allowances to 110 highly polluting, mostly coal-fired electricity-generating facilities. Facilities targeted under Phase I were required to achieve compliance with SO2 emission reductions by 1995. Phase II included smaller power-generating units that used oil, coal, and gas. The deadline for compliance with Phase II was January 1, 2000. I focus on the Phase I deadline because it targeted the largest coal-fired power plants, which were the largest customers for coal since the electric power industry accounted for 90 percent of coal demand in 2000 (Freme 2000). The ARP increased the demand for low-sulfur coal and reduced the demand for high-sulfur coal (U.S. Environmental Protection Agency 2001; Douglas and Wiggins 2015). Mines producing high-sulfur coal have a limited ability to post-process the coal to remove sulfur since part of the sulfur is integral to the chemical structure of the coal and cannot be removed (Wheelock and Markuszewski 1984).

Coal fired power plants were the main destination for U.S. coal production, so the national coal supply responded by reducing production of high sulfur coal while slightly increasing production of low sulfur coal. Figure 1 plots national coal production by sulfur content. Low sulfur coal production increases over this time period, though it remains relatively flat from 1994 until 2005. High sulfur coal production falls over the entire period and accelerates after 1990. The reduction in high sulfur coal production beginning around 1990, five years before the ARP deadline, reflects coal fired power plants switching coal suppliers in anticipation of the ARP deadline, since 70% of coal was supplied through contracts lasting between one and fifty years, while 15% of coal was purchased on the spot market, and 15% was sourced through vertically integrated coal mines and electric utilities (Joskow 1987).

4 Data

I measure the behavior of utilities and regulators under a pollution self-reporting regulation when pollution increases in a panel dataset of drinking water utilities downstream of at least one active coal mine from 1985 to 2005. Utility behavior and performance are measured through data on annual self-reporting violations, water quality violations, water pollution measurements, visits from the regulator, and received enforcement. I use a hydrological map to match coal mines to drinking water utilities with water intake downstream of a coal mine.

4.1 Utility characteristics, regulator, and violation data

The EPA's Safe Drinking Water Information System ("SDWIS") Enforcement and Compliance History Online ("ECHO") provides data on the universe of utilities active between 1985 and 2005. This time period lacks time-varying utility characteristic data, except for data on the number of facilities operated by each utility, such as water intakes and treatment plants. I use the number of active facilities as the only time-varying utility characteristic.

SDWIS ECHO provides data on utility MR violations, MCL violations, regulator enforcement, and utility visits. SDWIS records violations accurately but not exhaustively. A survey of 1,800 utilities in 2000 found that 10 percent of MR violations and 50 percent of MCL violations were in SDWIS, but 95 percent of violations in SDWIS were accurately reported (U.S. Environmental Protection Agency 2000).

I construct outcome variables for MR and MCL violations for a general IOC category, arsenic, and nitrates. SDWIS ECHO uniquely identifies utility violations with information on the start and end dates of the violation, the SDWA rule violated, and the pollutant. To avoid double counting violations, I exclude any IOC violations that occur at the same time and at the same utility as an arsenic or nitrate violation. I measure utility violations during a year with a binary variable that equals one if the utility experienced a violation of that type at any time during the year. I prefer this binary measure to the number of days per year a utility was in violation, which I summarize for comparison, because SDWIS ECHO often records violations as starting when utilities fail to meet regular testing schedules and then continue for a fixed period after they are issued, rather than recording the precise dates when pollution levels were elevated.

Table 1 presents summary statistics for two measurements of annual utility violations: whether there is any violation in a year and the number of days violated in a year. The summary statistics are calculated from utility-year cells. MCL violations rarely occur; a total of 49 MCL violations occur over the entire panel. MR violations occur roughly an order of magnitude more frequently than MCL violations; depending on the pollutant, between 402 and 612 MR violations occur during the panel. On average, utilities spend ten to fifteen days of the year in an MR violation; however, the distribution is highly bimodal since, when an MR violation occurs, it tends to last the entire year. The 90th percentile of the days in a year experiencing an MR violation is 0, and the 99th percentile is 365. The rarity of MCL violations results in a 99th percentile of zero days.

I measure regulator enforcement with a binary variable equal to one for every year a utility experiences enforcement. SDWIS ECHO enforcement records include the date enforcement began and ended, and whether it was formal or informal. The "Whole panel"

columns of Panel A of Table 2 summarize the share and number of cells in the sample where a utility received an enforcement action. Any enforcement occurs 20.8% of the time; most of that enforcement is informal, which occurs 18.2% of the time compared to 3% of the time for formal enforcement. A utility can experience both types of enforcement simultaneously, so "any enforcement" is not the sum of informal and formal enforcement. Enforcement is more likely to occur during a violation; conditional on an active violation, the rate of enforcement is 2 to 3.7 times higher than the unconditional rate. The "During MR year" and "During MCL year" columns of Panel A restrict the sample to utility-year observations where an MR or MCL violation of any type occurs. The likelihood of any enforcement increases from 20.8% in the unrestricted sample to 70.7% and 67.3% for the samples restricted to utility years with an MR and MCL violation, respectively.

I measure regulator visits to utilities with a binary variable, which is equal to one during a year when a utility received a regulator visit and zero otherwise. SDWIS ECHO records the date of the visit and reasons for visits, including sanitary survey, technical assistance, enforcement, sample collection, and inspections. The "Whole panel" columns of Panel B of Table 2 summarize the share and number of cells in the utility-year sample where a utility received a visit from the regulator. Sanitary survey visits are the most common type of regulatory visit, at 13.6%, compared to the second most common, inspection visits, at 3.4%. Technical assistance, enforcement, and inspection visits are generally more likely to occur during years when utilities receive an MR violation, at roughly twice the unconditional rate. Sample collection visits are less likely to occur during years with an MR violation compared to other years. This is counterintuitive since MR violations occur precisely when utilities do not self-report their water quality, and sampling information would reveal pollution levels. Sanitary survey visits occur at a similar rate, conditional on an MR violation, and at all times. This suggests that regulators conduct more activities than just sampling water quality in response to utility under-reporting. Regulators rarely use visits to respond to MCL violations. Visits rarely coincide with MCL violations; no more than five visits of any single type occurred during an MCL violation. I cannot draw a definitive conclusion given the infrequency of MCL violations.

4.2 Utility pollution concentration data

I measure pollution concentrations of arsenic, nitrates, barium, and selenium at utilities using data from the second Six-Year Review ("SYR2", sample years 1998–2005). Although this data only covers a subset of utilities and years, it allows me to test whether coal mining raises pollution levels at downstream utilities.

Beginning in 1991, the EPA conducted reviews of all National Primary Drinking Water Regulations every six years to determine whether the rules needed revisions. The SYR2 was the second round of the six-year review covering 1998 to 2005 and is the earliest six-year review round to be made publicly available. The SYR2 data contained pollution concentration data for 89 percent of the total population served by a drinking water system but excluded Pennsylvania, Mississippi, Kansas, and Louisiana because those states failed to submit records.

From the raw SYR2, I retain community water system utilities and drop transient and non-transient non-community water systems. This removes roughly 23 percent of raw monitoring records.

The data contains pollution concentration readings, whether the pollutant was detectable, and the detected concentration. Non-detections vary based on lab equipment sensitivity. To avoid a non-detect being interpreted as a pollution concentration of zero, I clean the SYR2 data following the procedure of Ravalli et al. (2022). For each pollutant, I compute the national share of records above the method detection limit ("MDL"), the minimum concentration at which a laboratory can reliably detect the pollutant in a sample, and I retain only pollutants detected in more than 10 percent of records. I replace the observations below the MDL of any pollutant with the MDL/$\sqrt{2}$; this prevents records from being dropped or mistakenly reported as zero without meaningfully inflating pollution levels. Records reporting a concentration greater than 100 times the pollutant's MCL are dropped as data-entry errors. Among the IOCs that can be produced during coal mining, this keeps arsenic, nitrate, barium, and selenium. All mass-concentration readings are converted to mg/L.

After cleaning, sample-level records are collapsed to a utility-by-pollutant-by-year panel. The annual pollution concentration for a particular pollutant is the mean of that year's samples at a particular utility.

Panel A of Table 3 summarizes the pollutants included in my panel. The maximum reading of every pollutant is below its MCL. The maximum arsenic reading of 0.0110 mg/L would exceed the stricter arsenic MCL of 0.010 mg/L, which took effect in 2006, but would not exceed the pre-2006 MCL of 0.05 mg/L. The fact that mean concentrations do not exceed the MCL is consistent with the low rates of MCL violations at utilities. Conditional on testing, pollution levels appear to be below the MCLs. These summary statistics do not provide information on the pollution concentrations that would be measured when a test was required by regulation but was not conducted by the utility. This data does not confirm that pollution levels never exceeded the MCL. Most MR violations were issued for failure to test water quality. The SYR2 necessarily reflects the results from

water quality tests and does not account for tests not undertaken that were subsequently punished with an MR violation.

SYR2 does not contain all test results submitted to regulators, but it is more likely to contain the test results of utilities that, at some point during my sample, fail to self-report their water quality and receive an MR violation. These results are more likely to reveal the pollution levels at the most negligent utilities since the SYR2 represents an oversampling of more negligent utilities that are more likely to have below average water quality. Utilities that appear in the SYR2 data are slightly more likely to receive an MR violation than those utilities that never appear in the data. Table 4 reports a two-sample t-test to compare the shares of utilities that appear in the SYR2 and have any SDWA violation to the share of utilities that never appear in the SYR2 data and have any SDWA violation. Utilities that appear in the SYR2 data are 7.2 percentage points more likely to experience any type of MR violation than utilities that never appear in the SYR2 data. The difference is significant at the 10% level. The lack of SYR2 observations for some utilities does not mean the utility was not self-reporting, since utilities can be compliant with MR rules and not have any SYR2 observations. The utilities that are slightly more likely to self-report data are also slightly less likely to have pollution concentration observations in SYR2.

4.3 River and stream connectivity data

Drinking water utilities are exposed to water pollution from coal mining when their source water is downstream of coal mines. The exposure is strongest when a utility draws from surface water. Groundwater sources downstream of coal mining are also exposed to upstream pollution since groundwater is recharged by water flowing from above. Groundwater sources are less exposed to pollution than surface water since polluted water flows through rock and substrate before recharging groundwater. I identify which utilities in the SDWIS ECHO national dataset source water from a watershed that is downstream of coal mining using the NHDplus hydrological connectivity dataset. The NHDplus hydrological dataset identifies the boundaries and connectivity of watersheds across the U.S.. I combine it with an EPA dataset on the watershed-level location of utility water intakes.

The NHDplus assigns watersheds a unique 2- to 12-digit code known as a hydrologic unit code (HUC). The number of digits corresponds to the size of the area covered by the watershed, with the geographical area decreasing with the number of digits; for example, the lower 48 U.S. states are divided into 18 HUC02s and over 87,000 HUC12s (see Fig-

ure 2). HUC12s in my sample are 11 km across on average. I use HUC12 watersheds to identify utility water intake locations. HUC12 watersheds are the smallest mapped watersheds available for the entire continental U.S. and are small enough that coal mining activities may impact the water in the HUC12 directly downstream. To ensure that a utility intake is downstream of a coal mine, I exclude those utilities with an intake located in the same HUC12 as a coal mine; a utility water intake at a HUC12's inflow may not be affected by mining at the same HUC12's outflow. I limit my sample to utilities with at least one water intake in a HUC12 at most two directly connected watersheds downstream from a HUC12 with a coal mine between 1985 and 2005.

4.4 Coal mine and coal sulfur data

I measure the number of coal mines and the tons of coal extracted in the watershed using individual coal mine production data from the Energy Information Administration ("EIA"). It includes the tons of coal produced by each mine in each year from 1985 to 2005. I merge EIA coal mine data with Mine Safety and Health Administration ("MSHA") data containing mine latitude and longitude coordinates. I match coal mines to watersheds from the NHDplus data.

My IV predicts a location's coal production from its geochemical properties and the preferences for those properties from coal fired power plants. In particular, coal fired power plants increased their preference for low sulfur coal following the ARP. I use the USGS COALQUAL dataset to measure the sulfur percentages present in coal within a watershed. The USGS COALQUAL dataset consists of geospatially explicit drilled coal core samples from across the U.S.. I match COALQUAL samples to watershed boundaries that lie within 20 km of their coordinates, irrespective of whether coal production took place in that watershed. This approximates coal deposits with the potential to be extracted. I assign a sulfur value of zero to watersheds that I cannot match to a drilled coal core sample.

Figure 3 maps sulfur as a percentage of coal's weight across watersheds for all coal sulfur samples. Sulfur levels vary continuously across and within regions and states. Neighboring watersheds have different sulfur levels. The variation in sulfur levels means predictions of coal production rely on more than just state, coal bed, or regional differences. The mismeasurement of sulfur levels due to extreme variation within watersheds, where some high-sulfur coal watersheds may actually contain low-sulfur coal, is not an issue since watersheds have low variation in measured coal sulfur levels.

The relative differences in coal production between high- and low-sulfur coal result

in within-region spatial variation regarding which watersheds experienced increases and decreases in production. Figure 4 plots the change in coal production at the watershed level between 1985 and 2005, with circles sized proportionately to the change. Within states and regions, coal production rarely experiences a uniform decline or increase. Coal production mostly decreased in Pennsylvania and mostly increased in West Virginia. Other states' coal production saw both increases and decreases. Production both increased and decreased in the western and eastern regions.

I match coal mines to utilities with water intakes within two watersheds downstream of a coal mine. I measure a utility's upstream coal production exposure as the sum of production across upstream watersheds. The coal production upstream of a utility depends on the average percentage of sulfur by weight in coal deposits in the watersheds where the utility's intakes are located and those that are upstream of the intakes. I measure a utility's upstream coal sulfur as the mean sulfur percentage across all watersheds within two watersheds upstream of its intakes.

In Figure 5, the watersheds that contain coal production and are upstream of a utility water intake are green, while the watersheds that contain coal production and are not upstream of a utility are red. Most watersheds in my sample are located in Pennsylvania, Kentucky, and West Virginia. The eastern coal fields more frequently occur in watersheds upstream of drinking water utilities. Western coal production takes place more frequently in watersheds without utility water intakes and is less well represented in my sample.

5 The impact of coal mining on pollution levels at downstream utilities

5.1 Empirical framework

I measure the effect of coal mining on IOC pollution concentrations at utilities. This establishes whether coal mining is a convincing source of pollution to study the effect of pollution on utility water quality self-reporting. I limit the estimation to arsenic, nitrates, barium, and selenium, as the MCL for these IOC pollutants did not change between 1985 and 2005, and these were the IOC pollutants that were retained after cleaning the SYR2 data.

Higher levels of coal production are associated with higher average pollution levels. Panel B of Table 3 measures the mean pollution concentration readings at utility-years with non-zero coal extraction in upstream watersheds. Compared to the mean pollution

concentrations in the full sample of utilities in Panel A, utility-years downstream of active coal extraction have slightly higher concentrations of arsenic and barium and slightly lower concentrations of nitrates and selenium. These raw comparisons do not account for utility and regional differences in pollution levels.

In order to control for utility and regional effects that influence pollution levels, I estimate the impact of coal mining on the mean pollution concentrations at utilities using the regression equation

$$Concen_{cht} = \beta \cdot \text{UpstreamCoalProd}_{ct} + \rho_c + HUC02_h \cdot \tau_t + \epsilon_{cht} \tag{3}$$

$Concen_{cht}$ is the mean pollution level of a pollutant measured at utility c in HUC02 h in the year t using SYR2 data from 1998 to 2005. $\text{UpstreamCoalProd}_{ct}$ is the coal produced in the year t, in millions of short tons, in the HUC12s upstream of utility c's intakes. β is the effect of an additional one million short tons of coal mined upstream of utility c in the year t. Utility fixed effects are included as ρ_c. To control for annual industrial, agricultural, and watershed level pollution trends that affected the entire watershed, I include HUC02 fixed effects interacted with time fixed effects $HUC02_h \cdot \tau_t$. Standard errors are clustered at the utility level.

Equation 3 identifies β from relative differences in exposure to annual coal production rather than comparing treated units to a control, since there are no utilities in my sample that are not downstream of coal production and can act as control observations. The identifying assumption requires that the pollution concentrations at utilities experiencing a large decline in coal production would have trended the same as the pollution concentrations at utilities with a small decline.

I do not treat these estimates as causal. There is no instrument for upstream production here and no untreated comparison group. The concentration data cover a subset of utilities from 1998 to 2005 rather than the full panel. The section establishes descriptive evidence for how coal production affects utility pollution concentrations. These results motivate using coal mining as a pollution shock to utilities.

5.2 Comparability of utilities by upstream production

Since the estimates of equation 3 are observational, it matters whether the utilities that received more upstream coal production differed from those that received less. Table 5 reports the coefficients from regressing upstream coal production from 1998 to 2005 on utility characteristics. I do not have utility characteristic data that varies over time, so I cannot include utility or year fixed effects. However, I include state fixed effects.

Columns one and two use annual production, the intensive margin from which equation 3 is identified, and columns three and four use an indicator for whether there was any upstream production.

On the intensive margin, utility characteristics are jointly insignificant with state fixed effects. Among utilities that have upstream mining, neither utility size, nor the number of water intakes a utility operates, nor whether it draws on surface water, nor whether it is publicly owned predicts how much coal is mined upstream. On the extensive margin, the characteristics are jointly significant, driven by the number of intake facilities: utilities operating more intakes are more likely to have any upstream mining at all, which is unsurprising since more intakes mean more watersheds in which a mine can sit. This motivates identifying equation 3 from the intensive margin of annual production rather than from whether a utility has any upstream production.

5.3 Results

Table 6 contains the results of equation 3 estimated from IOC pollution levels throughout utility water processing stages. The pollution concentration sample is small because it is restricted to the utilities and years for which the Six-Year Review reported readings.

Panel A restricts the sample to coal production in the watershed directly upstream of a utility intake. An additional one million short tons of coal mined in that year are associated with higher mean concentrations of arsenic and nitrates. Arsenic increases by 0.0034 mg/L with each additional one million short tons of coal mined upstream. This is economically significant, equivalent to 1.2 times the mean arsenic concentration of 0.0028 mg/L, and it is statistically significant at the 1% level. The effect represents 34 percent of the arsenic MCL of 0.010 mg/L that took effect in 2006, and 6.8 percent of the 0.050 mg/L MCL in force before then. An additional one million short tons of coal increase nitrates by 0.075 mg/L. The effect is statistically significant at the 10% level and accounts for almost 10 percent of the mean nitrate level.

Panel B restricts the sample to coal production within two watersheds upstream of a utility intake. An additional one million short tons of coal mined in that year increase nitrates by 0.061 mg/L. The effect is statistically significant at the 1% level. The effect on arsenic is no longer statistically or economically significant, consistent with arsenic not being carried as far downstream as nitrate.

I find that the estimates of coal production on pollution levels are robust to measuring coal production within two watersheds upstream, exactly two watersheds upstream, and two to three watersheds upstream. Appendix Figure A1 plots the effect of coal produc-

tion on pollution levels at different coal exposure definitions. The sample with exactly two watersheds upstream controls for concerns that a coal mine and a utility may be directly adjacent to each other but within two different watersheds. Coal mines and utilities that are directly adjacent to one another would likely make coordinated managerial decisions and undermine the separation between the utility and the source of pollution. I find that the point estimate is consistent across all watershed definitions, which supports the interpretation that coal mines and utilities are not making coordinated managerial decisions.

Coal production is associated with increases in pollution concentrations at utilities downstream. The effects are economically and statistically significant. This is evidence that upstream coal mining can be used to measure the response of utilities to increases in water pollution. Despite coal mining significantly increasing pollution, it is not associated with increases in pollution levels in the raw data above the MCLs.

6 An instrumental variable for upstream coal mining

I have shown that coal production is associated with higher water pollution levels at drinking water utilities. Two sources of endogeneity need to be addressed to identify the causal effect of upstream coal mining on utilities. First, coal mines are located in areas with particular geological characteristics, such as coal seam accessibility and transportation corridors, that may also independently predict water quality and utility characteristics. Second, utility characteristics are correlated with both upstream coal production and self-reporting behavior. A utility's performance is likely to change over the 20-year time span I study and cannot be controlled with utility fixed effects. Without accounting for utility characteristics that determine self-reporting and coal production leads to either a negative or positive omitted variable bias. For example, a poorly managed utility will under-report more often. A poorly managed utility may be less successful in lobbying regulators to prevent mines from opening in their water sources. This results in less well-managed utilities being exposed to more coal mines and under-reporting more. Without data on utility management, I would overestimate the impact of coal mining on utility under-reporting. If a regulator is interested in environmental justice it may prevent upstream coal mines from opening in source water used by poorly performing utilities.[3] Low coal production would be associated with poorly performing utilities that self-report

3. The EPA Source Water Protection program can designate areas where drinking water is sourced, especially for utilities with poor water quality performance, and take an inventory of pollution from human activities or natural sources that pose the greatest threat to source water, making it difficult to site pollution-producing activities in these areas.

less often. Without accounting for utility performance, this would suggest that more coal production is associated with improvements in self-reporting.

I adapt an instrument from Douglas and Wiggins (2015) to predict drinking water utilities' exposure to coal mining. The instrumental variable measures changes in coal production that resulted from national trends in coal demand and were unrelated to a utility's characteristics. Coal-fired power plants are the main consumers of coal and have preferences regarding the chemical properties of the coal they use. In 1995, coal-fired power plants were required to meet the sulfur dioxide emission restrictions of the ARP and the demand for coal containing a lower amount of sulfur increased. Coal production in a watershed depends partly on the sulfur content of coal in the watershed and whether the time period is before or after 1995.

I estimate a two-stage least squares regression. The first stage is a regression of the number of active coal mines upstream of a utility on the mean sulfur percentage of coal weight across its upstream watersheds, interacted with a binary variable equal to one for all years after 1995 and zero otherwise. The first stage is

$$\begin{aligned}\text{UpstreamCoalMines}_{ct} = \alpha_0 + \alpha_1 \text{NumIntakeFacilities}_{ct} + \alpha_2(\text{Post1995}_t \times \text{UpstreamSulfur\%}_c) \\ + S_s \times Y_t + C_c + \nu_{ct}\end{aligned} \tag{4}$$

The dependent variable, $\text{UpstreamCoalMines}_{ct}$, is the number of active coal mines in watersheds within two watersheds upstream of utility c's intakes in year t, summed across those watersheds. The exogenous covariates are $\text{NumIntakeFacilities}_{ct}$, the number of intake facilities the utility operates in year t, and utility fixed effects C_c. In the preferred specification, I include state-by-year fixed effects $S_s \times Y_t$, which absorb shocks common to all utilities in a state in a given year. AMD is more common when the minerals contain high levels of sulfur, making the incidence of AMD increase with coal sulfur levels (Schweinfurth 2009). Upstream sulfur is controlled for with utility fixed effects since sulfur levels do not change over time. The excluded instrument is the interaction between $\text{UpstreamSulfur\%}_c$, the mean sulfur percentage of coal weight across the watersheds upstream of utility c, and Post1995_t, a dummy variable equal to one for the years after 1995, the deadline to implement Phase I of the ARP, and zero otherwise. Standard errors are clustered at the utility level.

The coal sulfur content before and after 1995 predicts the amount of coal production well. Figure 6 plots the mean and standard deviation of HUC12 coal sulfur percentages by bins of the number of active coal mines in a HUC12 before 1995 and after 1995. In

addition, it plots linear fit lines separately for before 1995 and after 1995. After 1995, coal production heavily shifted towards coal with sulfur percentages below 1.5%. The change in production away from high-sulfur coal after the ARP is strongly demonstrated by the line of linear fit becoming more downward sloping after 1995, indicating that while there was a bias towards coal production in low-sulfur coal areas before 1995, there was a strong increase in preference away from high-sulfur coal or towards even lower sulfur coal after 1995. The time-contingent switch in coal production based on sulfur content supports the structure of the variable $\text{Post1995}_t \times \text{UpstreamSulfur\%}_c$.

The strength of the instrument is still evident after controlling for covariates and fixed effects. Figure 7 is a scatter plot of the residualized number of upstream coal mines and the residualized instrumental variable. There is a negative relationship between the residualized number of upstream coal mines and $\text{Post1995}_t \times \text{UpstreamSulfur\%}_c$.

Table 7 reports the first-stage F-statistic, computed as the squared clustered t-statistic on the excluded instrument, which equals the effective F-statistic of Montiel Olea and Pflueger (2013) in this just-identified setting. The F-statistic of 49.2 exceeds their critical value of 23.1 for a 5%-level test against 10% worst-case bias, well above the conventional rule-of-thumb threshold of 10 proposed by Staiger and Stock (1997).

The instrument requires monotonicity in the first stage to recover the local average treatment effect (LATE) estimate of the instrument.[4] In this setting, the instrument is $Z_c = \text{Post1995}_t \times \text{UpstreamSulfur\%}_c$, which equals zero before 1995 and $\text{UpstreamSulfur\%}_c$ after. For a given sulfur level w, monotonicity requires either $\text{UpstreamCoalMines}_c(0) \leq \text{UpstreamCoalMines}_c(w)$ for all c, or $\text{UpstreamCoalMines}_c(0) \geq \text{UpstreamCoalMines}_c(w)$ for all c. This permits the direction of the response to vary with sulfur content. For example, at $\text{UpstreamSulfur\%}_c = 1$ (low sulfur), the number of mines increases after 1995, so monotonicity requires $\text{UpstreamCoalMines}_c(0) \leq \text{UpstreamCoalMines}_c(1)$ for all c. At $\text{UpstreamSulfur\%}_c = 4$ (high sulfur), the number of mines falls after 1995, so monotonicity requires $\text{UpstreamCoalMines}_c(0) \geq \text{UpstreamCoalMines}_c(4)$ for all c. These two conditions are compatible: monotonicity is satisfied so long as each pairwise comparison is uniform across units, even when low- and high-sulfur areas respond in opposite directions.

An implication of monotonicity is that there are no defier utilities. A defier utility experiences an increase in the number of coal mines it is downstream of after 1995 due to being downstream of a high-sulfur coal watershed, but it would experience a decrease

4. Stating monotonicity more formally and following Angrist and Imbens (1994), let Ψ be the support of the instrument Z_c, and for each $z \in \Psi$ define a random variable $D_c(z)$. The monotonicity assumption requires that for all $z, w \in \Psi$, either $D_c(z) \leq D_c(w)$ for all c, or $D_c(z) \geq D_c(w)$ for all c. This condition applies pairwise to elements of Ψ and imposes no restriction on the ordering of z and w.

in the number of coal mines it is downstream of after 1995 due to being downstream of a low-sulfur coal watershed. This is unlikely since high-sulfur coal was more expensive to operate than low-sulfur coal after 1995. To be a defier requires a power plant that consumes high-sulfur coal from a watershed to increase its coal demand after installing sulfur scrubbers and to increase its electricity market share. Power plants that rely on high-sulfur coal and scrubbers are likely to demand less coal since scrubbers are expensive to install, increase operating costs, require higher electricity rates to retain profits, and lose electricity market share. If some high-sulfur regions had a comparative advantage in low-cost production that offset the ARP cost increase, such as very low labor costs or transportation advantages, they might expand production even as ARP made their coal relatively more expensive. Such an expansion creates defiers only if it would not have occurred had those same low-cost regions instead contained low-sulfur coal; otherwise, it creates never-takers. A never-taker is a utility that is downstream of a low-sulfur watershed that experiences no increase in coal mines after 1995, and a utility that is downstream of a high-sulfur watershed that experiences no decrease in coal mines after 1995. A never-taker is not a violation of monotonicity and only reduces first-stage power.

The second stage of the IV regression is

$$\begin{aligned} \text{UtilityOutcome}_{ct} = \beta_0 + \beta_1 \text{NumIntakeFacilities}_{ct} + \gamma \widehat{\text{UpstreamCoalMines}}_{ct} \\ + S_s \times Y_t + \epsilon_{ct} \end{aligned} \tag{5}$$

In the second stage, the dependent variable $\text{UtilityOutcome}_{ct}$ is a binary variable equal to one if utility c received an MR violation, MCL violation, regulator visit, or enforcement from the regulator in year t. Since the sulfur content of a coal deposit is a fixed geological endowment, I control for it with utility fixed effects. In each results table, the 2SLS row "Upstream coal mines" reports γ, the OLS row reports the coefficient on $\text{UpstreamCoalMines}_{ct}$ when equation 5 is estimated without instrumenting, and the reduced-form (RF) row reports the coefficient from regressing the outcome directly on $\text{Post1995}_t \times \text{UpstreamSulfur\%}_c$.

In addition to instrument relevance, the IV produces causal estimates of the impact of coal mining on utility and regulator behavior if the instrument satisfies the independence and exclusion restrictions (Angrist and Pischke 2009). Conditional on utility and state-by-year fixed effects, the changes to upstream coal production to which utilities are exposed through the instrument are independent of their potential outcomes because the scope of the ARP and the timing of its first phase were both determined independently of utility drinking water quality and utility characteristics. The instrument identifies variation in

the change in coal production that is independent of unobserved utility characteristics and is based on the shift in coal-fired power plants' demand for lower-sulfur coal after 1995 in order to meet the sulfur dioxide emission standards of the ARP. The ARP was a national clean air policy that was not written or enacted to address drinking water quality issues. The instrument exploits the national change in power plants' demand for coal by sulfur content, the timing and stringency of which were set without reference to any drinking water utility. The effects of the ARP on coal production were determined by national and international energy markets that local drinking water utilities or local utility drinking water quality did not affect. Local utility characteristics regarding drinking water quality or local power over source protection did not affect the scope of the ARP; the stringency of the ARP law; the degree to which low-sulfur coal should be part of the abatement technology; or the deadline power plants were required to meet the first stage emission requirements. In addition, since the timing of the ARP did not coincide with major drinking water regulations the instrument does not spuriously correlate with laws that would be important for utility behavior.

The instrument would violate independence if the unobserved utility characteristics that affect drinking water quality violations and the siting of coal mines changed after 1995 for utilities with different amounts of sulfur as a percentage of coal weight. Knowing about sulfur as a percentage of coal weight upstream of a utility after 1995 tells us nothing about how much political power a local utility has over mine siting.

Exogeneity requires the instrument to affect utilities only through its effect on the pollution from upstream coal mines. A threat to this would be if the SDWA regulation changed after 1995 and had a differential impact on utilities downstream of high-sulfur coal before and after 1995 compared to utilities downstream of low-sulfur coal before and after 1995. Changes in SDWA rules that affected all utilities in the same way are controlled for with state-by-year fixed effects. For example, in 1991, the monitoring and reporting schedule changed for all utilities and was controlled for with fixed effects. Any change to an MCL would apply to all utilities and be controlled for with time fixed effects. Any change to a rule or enforcement that only affected one state are also controlled for by the state-by-year fixed effects.

The exclusion restriction fails when the instrument affects regulator behavior through another path outside its effect on coal mining. The exclusion restriction would be threatened if the ARP Phase I directly changed state EPA enforcement practices in a way that affected utilities differently according to the percentage of sulfur by weight in upstream coal. However, states were not primarily responsible for enforcing the ARP; it was a federally regulated program that, by itself, would not have affected state primacy enforcement

of the SDWA.

Another exogeneity threat is local economic activity. The shift in demand away from high-sulfur coal reduced mining employment and income in the communities that a utility serves, which could affect a utility's staffing and revenue, as well as a state regulator's enforcement priorities, without operating through water pollution. State-by-year fixed effects absorb this channel to the extent that it operates at the state level, so I report every main specification both with year fixed effects and with state-by-year fixed effects.

The exclusion restriction cannot be tested directly in a just-identified model, but individual alternative pathways can be tested. I test one such pathway by regressing the number of a utility's active intake facilities, a utility characteristic that varies over time, on the instrument. This variable also enters both stages as an exogenous covariate, so evidence that the instrument moved it would indicate both an alternative pathway and that it is an inappropriate control. Table 8 reports the results. The coefficient on the instrument is small and statistically indistinguishable from zero, both in the specification with utility and state-by-year fixed effects and in the specification with year fixed effects on a balanced panel. This rules out one pathway but does not prove the exclusion restriction since that is fundamentally untestable.

The IV exogenously predicts upstream coal production from the geochemistry of the coal upstream and the shift in demand for low sulfur coal. If coal geochemistry from a downstream coal bed predicts upstream coal production, then the instrument is capturing a feature of coal production that is common in a region, such as local supply chains, which may also be correlated with utility characteristics. As a robustness test, I predict upstream coal production with my preferred instrument $\text{Post1995}_t \times \text{UpstreamSulfur\%}_c$ and a placebo instrument $\text{Post1995}_t \times \text{DownstreamSulfur\%}_c$, where $\text{DownstreamSulfur\%}_c$ is the mean percentage of sulfur by coal weight downstream of a utility intake. This is an over-identified instrument for a utility's exposure to upstream coal production with $Z_c = [\text{Post1995}_t \times \text{UpstreamSulfur\%}_c,\ \text{Post1995}_t \times \text{DownstreamSulfur\%}_c]$. Table 15 in Appendix A.2.2 contains the estimates from the first stage with the over-identified instrument. On its own, the downstream instrument predicts upstream coal mines, but it does not explain the number of coal mines upstream of a utility once $\text{Post1995}_t \times \text{UpstreamSulfur\%}_c$ is included. With both instruments included, the coefficient on the upstream instrument has the same magnitude and significance as in the preferred specification, suggesting that the placebo instrument's predictive power comes from its correlation with upstream sulfur rather than from a shared regional feature of coal production.

The over-identified instrument does not predict utility characteristics (Table 16); a relationship would have suggested that the instrument was not exogenous.

Relevance, independence, and exclusion are sufficient for the IV estimates to produce causal estimates of the effect of coal mining on drinking water utility and regulator behavior. Relevance is verified above. Independence and exclusion are not testable; I argue for each on the institutional grounds set out in this section. Monotonicity is required not for causality but for interpretation. Under monotonicity, the estimate is a local average treatment effect: the effect of an additional upstream coal mine at the utilities whose upstream mining responded to the ARP. Were monotonicity to fail, the estimate would still be assembled from causal effects, but it would weight defier utilities negatively and would no longer be an average effect for an identifiable group of utilities.

7 The impact of coal mining on utility under-reporting and water quality violations

I now measure the effect of coal mining on utility self-reporting, measured by MR violations, and on MCL violations. The theoretical model predicts that coal mining causes utility self-reporting to fall and the probability of an MCL violation to increase. Table 9 reports the results of equation 5, estimating the effect of an additional active coal mine on the likelihood that a utility receives any violation, either an MR or an MCL violation, for nitrates, arsenic, and inorganic chemicals using OLS, two-stage least squares, and reduced-form regressions.

Coal mining increases the likelihood that a utility experiences an MR or MCL violation in specifications with utility and state by year fixed effects and those with utility and year fixed effects. Estimated with OLS and utility and state-by-year fixed effects, an additional active coal mine upstream increases the likelihood of an MCL or MR violation for nitrates, arsenic, and IOCs by 0.54, 0.43, and 0.28 percentage points, respectively. The reduced form effect of the instrument measures how much more a one-percentage-point increase in upstream sulfur affects the likelihood of violations after ARP Phase I relative to before. A one-percentage-point increase in upstream coal sulfur leads to a 1.5 to 2 percentage point reduction in the likelihood that a utility experiences an MCL or MR violation for nitrates, arsenic, or IOCs after 1995 compared to before 1995. Since coal production fell most acutely in high sulfur coal areas after 1995, this suggests that in areas where the number of coal mines decreases and upstream pollution subsequently falls, the likelihood of a violation decreases.

In the IV regression, one additional active coal mine increases MCL or MR violations for nitrates, arsenic, and IOCs by 3.89, 4.00, and 3.09 percentage points, respectively.

These effects are statistically significant at the 10% level for IOCs and at the 5% level for arsenic and nitrates. The OLS estimates are roughly one-seventh to one-eleventh the size of the IV estimates, suggesting that the omitted variable bias is negative.

Coal mining raises MR violations, supporting the theoretical model's Proposition 2. The results presented in Table 10 estimate equation 5 on MR violations of arsenic, nitrate, and IOCs. An additional coal mine increases the likelihood of an MR violation for nitrates, arsenic, and IOCs by 3.94, 3.78, and 3.06 percentage points, respectively. These effects are of the same magnitude as the effects of coal mining on a combined measure of MR and MCL violations, so the entire effect of mining on utilities is driven by MR violations rather than MCL violations. The effects on MR violations for arsenic and nitrates are estimated with the same precision as the effects on any violation for those pollutants. The effects range from roughly 70 percent of the mean violation rate to the full mean rate. Table 1 reports the mean MR violation rates of arsenic, nitrates, and IOCs at 3.73 percent, 5.68 percent, and 4.31 percent, respectively.

Coal mining has almost no economically significant effect on the incidence of MCL violations. The impact of coal mining on utility MCL violations for nitrates, arsenic, and IOCs is presented in Table 11. Arsenic MCL violations are 0.22 percentage points more likely to occur with an upstream coal mine instrumented with the ARP timing and upstream coal sulfur. This effect is statistically significant at the 5% level. The IV estimated effects of coal mines on the likelihood that a utility experiences a nitrate or inorganic chemical MCL violation are statistically and economically insignificant. The point estimate for the effect of an additional mine on the likelihood that a utility receives a nitrate MCL violation is -0.11 percentage points, and for an inorganic chemical MCL, it is -0.20 percentage points. Neither of these effects is statistically significant, reflecting the extreme rarity of MCL violations in the sample.

The main results measure upstream coal mines within two watersheds upstream of a utility intake. These results are robust to measuring upstream coal mines two to three watersheds and exactly two watersheds away (Appendix Figure A2).

I measured the relationship between coal mining and water pollution in Section 5 using tons of coal produced upstream and measured the relationship between coal mining and utility behavior using the number of active coal mines upstream. As a robustness test that the results do not depend on which definition of upstream coal mining I use, I divide the outcome and explanatory variables of equation 5 by their standard deviations and, separately, by their sample means. This gives results in standard deviation units and elasticities at the sample means. Appendix Figure A4 reports the first stage for both definitions, and Appendix Figure A5 reports the effects on MR and MCL violations. The

effect of coal mining on utility behavior is robust to either definition of upstream coal mining.

The IV results are robust to including a placebo instrument that should not predict upstream coal production, $\text{Post1995}_t \times \text{DownstreamSulfur\%}_c$, reported in Appendix A.2.2 in Tables 17–19.

Rising MR violations are consistent with Proposition 2 in the theoretical model. Regulators require more self-reporting from utilities downstream of impaired water, raising self-reporting costs. The theoretical model predicts that mining raises under-reporting through a second channel by changing the compliance costs. Compliance costs rise with mining because mining raises baseline pollution levels and the effort required to treat water to any given standard. As coal mining increases, self-reporting costs and compliance costs rise, both of which increase the benefits of under-reporting.

Coal mining would not affect MCL violations if it did not raise pollution levels above the MCL in my sample, or if MR violations concealed MCL violations. In the theoretical model, concealed MCL violations are discovered by the regulator through inspections of under-reporting utilities.

8 The impact of coal mining on regulator visits and enforcement

I estimate the response of regulator visits to coal mining, including sanitary surveys, enforcement visits, and inspections. Table 12 presents the response of regulator visits to coal mining by estimating OLS, equation 5, and the reduced form equations. Column five reports the response of inspection visits. I find that an additional coal mine leads to a utility experiencing a 2.13 percentage point increase in inspection visits by regulators. This is statistically significant at the 5% level, and since it represents a 62% increase over the baseline inspection visit rate of 3.43%, it is economically significant. Regulators are likely to discover concealed MCL violations during inspection visits and issue punitive or corrective enforcement. Column three reports the response of enforcement visits to coal mining; an additional coal mine raises the likelihood of an enforcement visit by 1.07 percentage points, which is not statistically significant. The effects on sanitary, technical assistance, and sample collection visits are also not statistically significant.

I estimate the effect of coal mining on the enforcement utilities receive. This provides evidence of the results of regulator inspections since regulators would punish utilities harshly for under-reporting in order to conceal an MCL violation. This also provides evi-

dence for the disincentives regulators create for utility under-reporting. Table 13 presents the responses of informal enforcement, formal enforcement, and no enforcement to coal mining by estimating OLS, equation 5, and a reduced form. I cannot detect a statistically significant effect of coal mining on enforcement when I include state-by-year fixed effects. The point estimates for informal and formal enforcement are negative, suggesting that coal mining does not increase the likelihood that a utility receives enforcement. This evidence suggests that coal mining is unlikely to cause utilities to under-report in order to conceal MCL violations, since despite an increase in inspection visits, enforcement does not increase. If utilities were concealing MCL violations, the higher rate of inspection visits caused by mining would likely uncover them, and enforcement would increase. The enforcement results also suggest that the penalties for under-reporting are mild, indicating that utilities have a low regulatory disincentive to under-report. A low penalty for under-reporting, t in the theoretical model, lowers the self-reporting cutoff $\hat{c}(a)$ at every level of negligence, regardless of the probability of inspection and the sanction for a water quality violation.

The main results measure upstream coal mines within two watersheds upstream of a utility intake. These results are robust to measuring coal mining within two to three watersheds upstream and within exactly two watersheds (Appendix Figure A3).

I measure the relationship between coal mining and water pollution in Section 5 using tons of coal produced upstream, and I measure the relationship between coal mining and regulatory and enforcement behavior using the number of active coal mines upstream. As a robustness test to ensure that the results do not depend on which definition of upstream coal mining I use, I convert the variables in equation 5 into standard deviation units by dividing by the standard deviation and into elasticities at the sample mean units by dividing by their sample means. I report these results in Appendix Figure A6. Converting to standard deviation units and elasticities at the sample mean does not affect the coefficient estimates or significance. The effect of coal mining on regulator visits and enforcement is robust to either definition of upstream coal mining.

The IV results are robust to including a placebo instrument that should not predict upstream coal production, $\text{Post1995}_t \times \text{DownstreamSulfur\%}_c$, reported in Appendix A.2.2 in Tables 20–21.

9 Discussion

By raising self-reporting costs, coal mining can lower self-reporting even if it does not push pollution above the MCL. Pollution levels do not rise above the MCL at utilities in

the SYR2. These utilities differ in their compliance costs and, returning to the model, in their negligence. This implies that over the range of utilities in the data, the likelihood of exceeding the MCL upon self-reporting is small, $q(a) \to \epsilon$. The self-reporting cutoff $\hat{c}(a) = t - (r - ps)q(a)$ then sits close to t, and a utility with no MCL violation to conceal decides whether to report by weighing the direct cost of self-reporting against the penalty for under-reporting. Since self-reporting costs are enough to reduce self-reporting, making the MCL stricter would not improve it.

I find evidence that enforcement does not increase for utilities that under-report due to coal mining pollution. This reduces the disincentive to under-report due to increases in self-reporting costs.

I find evidence that under-reporting is primarily in response to self-reporting costs and not to concealing water quality violations. The SYR2 pollution concentrations are far below the MCL. A utility would not exceed the MCL for barium, nitrates, or selenium, even at the maximum pollutant level observed in the sample and the highest single-year dose of upstream coal production. Arsenic is the exception: the sample maximum of 0.011 mg/L remains below the pre-2006 MCL of 0.05 mg/L but exceeds the stricter 0.01 mg/L MCL that took effect in 2006. This suggests that utilities maintain water quality well below MCL levels. However, this result is limited to a subset of utilities from 1998 to 2005.

I test if self-reporting costs alone induce under-reporting by estimating the effect of SYR2 pollutant concentrations of nitrates on the likelihood of nitrate self-reporting violations. A utility is required to increase nitrate testing from once a year to four times a year when it detects nitrates above 50% of the MCL.[5] While nitrate pollution levels remain above 50% of the MCL and below the MCL, a utility experiences an increase in self-reporting costs without a water quality violation. By estimating the likelihood of a utility under-reporting nitrates when nitrates are above 50% and below 100% of the MCL, I test if self-reporting costs alone reduce self-reporting. Table 14 presents the results; columns one and two report the likelihood of an MR violation for nitrates within one year and within six months of a nitrate pollution reading. A nitrate reading above 50% of the nitrate MCL and below 100% of the MCL increases the likelihood of a nitrate self-reporting violation by 59 percentage points and 26 percentage points within the next twelve and six months, respectively. I find no effect on the likelihood of an MR violation for nitrates in the twelve or six months following a one standard deviation increase in mean nitrate pollution measurement. These effects, which are not causal, suggest that self-reporting

5. Other IOCs increase testing based on source water risk instead of a single pollution concentration threshold.

rates decrease as self-reporting costs increase.

Finally, regulators do not issue enforcement actions to utilities that are consistent with utilities concealing water quality violations. I find evidence that coal mining pollution causes regulators to increase inspection visits to utilities to determine water quality and, ultimately, the health risk of an under-reporting utility. Despite the increase in inspections, those same utilities do not receive increases in enforcement actions. Inspections would likely uncover concealed water quality violations. The combined evidence that coal mining causes regulators to conduct more inspection visits while not increasing enforcement suggests that utilities are not concealing MCL violations.

I do not address the welfare effects of shifting monitoring onto the regulators. Water quality is generally perceived as a local issue with limited externalities (Olmstead 2010). Without externalities, it may be inefficient for a state regulator to expend resources to monitor a local utility. Small utilities find the burden of water quality management particularly heavy. Regulators may benefit from economies of scale that make their monitoring cheaper than the utilities' cost of self-reporting. However, the economies of scale would have to overcome the large amount of resources that the regulator would have to devote to monitoring all utilities in a timely manner. To measure efficiency, it is necessary to take into account the regulators' preferences, which affect enforcement, not only violation severity (Kang and Silveira 2021).

The theoretical model suggests that under-reporting can be reduced during times of increasing pollution by raising the penalty for under-reporting or by reducing the cost of self-reporting. Continuous monitoring technology that the regulator controls could reduce the cost of self-reporting and make violations easier to spot, making successful enforcement more likely.

10 Conclusion

I present evidence that, under a mandatory self-reporting regulatory framework, self-reporting falls when costs increase, even when the regulated are otherwise in compliance with the regulation. I show this in the context of coal mining and utilities. Utility self-reporting of water quality falls when water pollution from coal mining increases. An additional active coal mine upstream raises the likelihood of under-reporting water quality for inorganic chemicals, arsenic, and nitrates by three to four percentage points. Utility under-reporting increases when coal mining impairs water quality at utilities and raises the amount of self-reporting that utilities are required to complete. Consistent with the cost channel, a nitrate reading above half the legal limit, the level at which the regula-

tor requires utilities to increase nitrate testing from once a year to four times a year, is followed by a large increase in the likelihood of failing to submit a self-reported water quality test for nitrate. For the subset of utilities for which I observe water pollution levels, I do not detect pollution levels rising above legal limits.

I find that penalties for not self-reporting are not high enough to entirely offset the utility's incentive to under-report when self-reporting costs rise. Regulators respond to upstream mining by diverting their attention to under-reporting utilities rather than sanctioning them. An additional upstream mine raises the likelihood of an inspection visit by 2.1 percentage points, while having no detectable effect on enforcement. The fact that regulators inspect these utilities more often without enforcement suggests that the visits find nothing to sanction and that utilities are not under-reporting to conceal a water quality violation. The regulator can use inspection visits to assess the threat posed to public health by a missing water quality report and can implement a less costly type of enforcement when the threat is discovered to be low. However, without punishing under-reporting, regulators will have to conduct more inspection visits to ensure that pollution levels do not exceed the legal limit.

The findings here suggest that the regulatory design of the SDWA, which relies on utility self-reporting, assigns low penalties for under-reporting, and requires likely water quality violators to conduct more self-reporting, creates systematic incentives to under-report. The theoretical model suggests that regulators can increase self-reporting by increasing penalties for under-reporting or reducing the costs of self-reporting.

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Tables and Figures

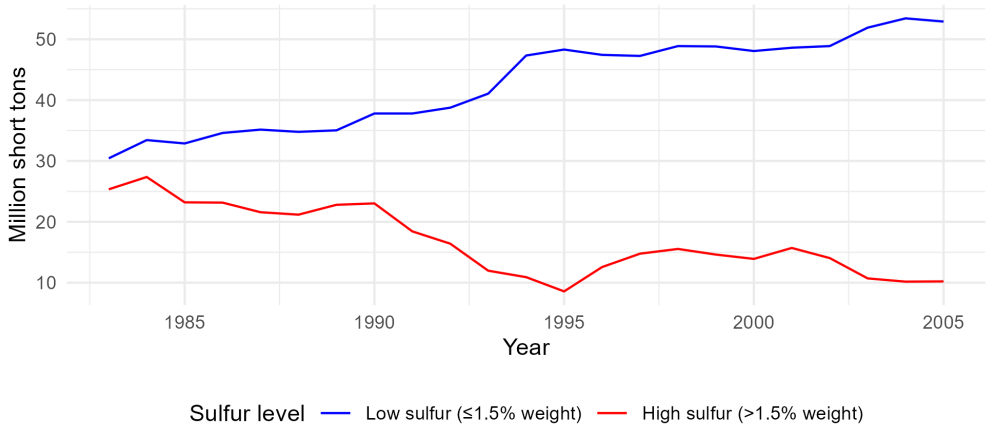


Figure 1: Coal production trends by percent of sulfur in coal by weight, 1983 to 2005.

Table 1: Inorganic chemical violations at utilities, 1985–2005

Panel A: Any violation in year

	MR		MCL	
Pollutant	**%**	**N**	**%**	**N**
Nitrates	5.68	612	0.11	12
Arsenic	3.73	402	0.05	5
Inorganic chemicals	4.31	465	0.30	32

Panel B: Days in year

	MR				MCL			
Pollutant	**Mean**	**SD**	**P90**	**P99**	**Mean**	**SD**	**P90**	**P99**
Nitrates	14.54	65.35	0.00	365.00	0.27	9.19	0.00	0.00
Arsenic	9.98	54.32	0.00	365.00	0.08	3.83	0.00	0.00
Inorganic chemicals	11.53	58.47	0.00	365.00	0.71	15.08	0.00	0.00

Notes: Sample of drinking water utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, 1985–2005. MR = monitoring and reporting violation; MCL = maximum contaminant level violation. Number of observations = Number of utilities $\times$ Number of years. N = 10,782 = 582 utilities $\times$ up to 21 years (1985–2005).

Table 2: Enforcement actions and site visits at utilities, 1985–2005

	Whole panel		During MR year		During MCL year	
	%	N	%	N	%	N
Panel A: Enforcement type						
Formal	3.00	323	9.41	70	6.12	3
Informal	18.24	1,965	66.67	496	59.18	29
Any	20.75	2,235	70.70	526	67.35	33
Panel B: Visit type						
Sanitary	13.59	1,464	13.17	98	10.20	5
Technical assistance	2.29	247	4.70	35	2.04	1
Enforcement	1.05	113	2.02	15	4.08	2
Sample collection	0.78	84	0.54	4	0.00	0
Inspection	3.43	369	5.65	42	2.04	1

Notes: Sample of drinking water utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, 1985–2005. MR = monitoring and reporting violation; MCL = maximum contaminant level violation. During MR year and During MCL year columns restrict to the 744 and 49 cells, respectively, with a nitrate, arsenic, or inorganic chemical violation of that category, before computing the same share and count. Panel A: Formal and Informal enforcement follow the EPA SDWIS enforcement-action classification; Any is a cell with an enforcement action of either category. Panel B: Sanitary visits are sanitary surveys and follow-up sanitary surveys; Technical assistance covers technical assistance, engineering, and operations and maintenance visits; Enforcement visits are formal-enforcement, investigation, and emergency assistance visits; Sample collection is a sample-collection visit; Inspection covers site, regularly scheduled, and informal system inspections. An enforcement action or visit type may co-occur with others in the same cell, so rows need not sum to the Any/whole-panel total. Number of observations = 10,773.

Table 3: Utility pollution concentrations, 1998–2005

Variable	MCL	Mean	Max	Std. dev.	*N*	Near MCL (%)
Panel A: Full sample						
Arsenic (mg/L)	0.050 (pre-2006), 0.010 (2006+)	0.0028	0.011	0.0023	416	0.00
Nitrate (mg/L)	10	0.80	5.5	0.87	488	0.41
Barium (mg/L)	2.0	0.074	0.71	0.066	407	0.00
Selenium (mg/L)	0.050	0.0047	0.035	0.0049	405	0.49
Panel B: Utility-years with non-zero coal extraction						
Arsenic (mg/L)	0.050 (pre-2006), 0.010 (2006+)	0.0030	0.010	0.0024	93	0.00
Nitrate (mg/L)	10	0.73	4.0	0.67	118	0.00
Barium (mg/L)	2.0	0.076	0.21	0.048	91	0.00
Selenium (mg/L)	0.050	0.0044	0.035	0.0048	90	1.11

Notes: SYR2 sample from 1998–2005. "Near MCL" is the percent of utility-year mean concentrations exceeding 50% of the applicable MCL. Panel B restricts Panel A's sample to utility-year observations with non-zero annual coal extraction in upstream watersheds.

Table 4: MR and MCL violation rates and counts by SYR2 pollution concentration reporting status, two-step upstream watershed linkage, 1985–2005

	Utility appears in SYR2 (%)			
	Yes (1)	No (2)	Diff. (3)	*p*-value (4)
Ever MR violation	51.27	44.08	7.19*	0.09
Ever MCL violation	5.51	3.55	1.96	0.28
Mean annual violations (count, 0–6)	0.13	0.20	-0.07**	0.01

Notes: Sample restricted to utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake. *Has SYR2 reading*: at least one Six-Year Review pollution concentration reading for arsenic, nitrate, selenium, or barium. *No SYR2 reading*: utility is in a state where at least one utility has a SYR2 reading but the utility itself does not. MR violation: inorganic chemical monitoring and reporting violation. MCL violation: inorganic chemical maximum contaminant level violation. Mean annual violations: sum of MR and MCL violation indicators across arsenic, nitrate, and inorganic chemicals categories (maximum of 6 per utility-year). *N* utilities: group 1 = 236, group 2 = 338. The first two rows are percentages of utilities and their differences are in percentage points. Stars on differences are from two-sample *t*-tests. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.



Figure 2: Hierarchical structure of USGS hydrologic unit codes. Illustrates how a hydrologic region is nested into progressively smaller watershed units, from two-digit units covering large multi-state regions down to twelve-digit units covering a few square miles. Each smaller unit is fully contained within the boundaries of the larger unit above it.

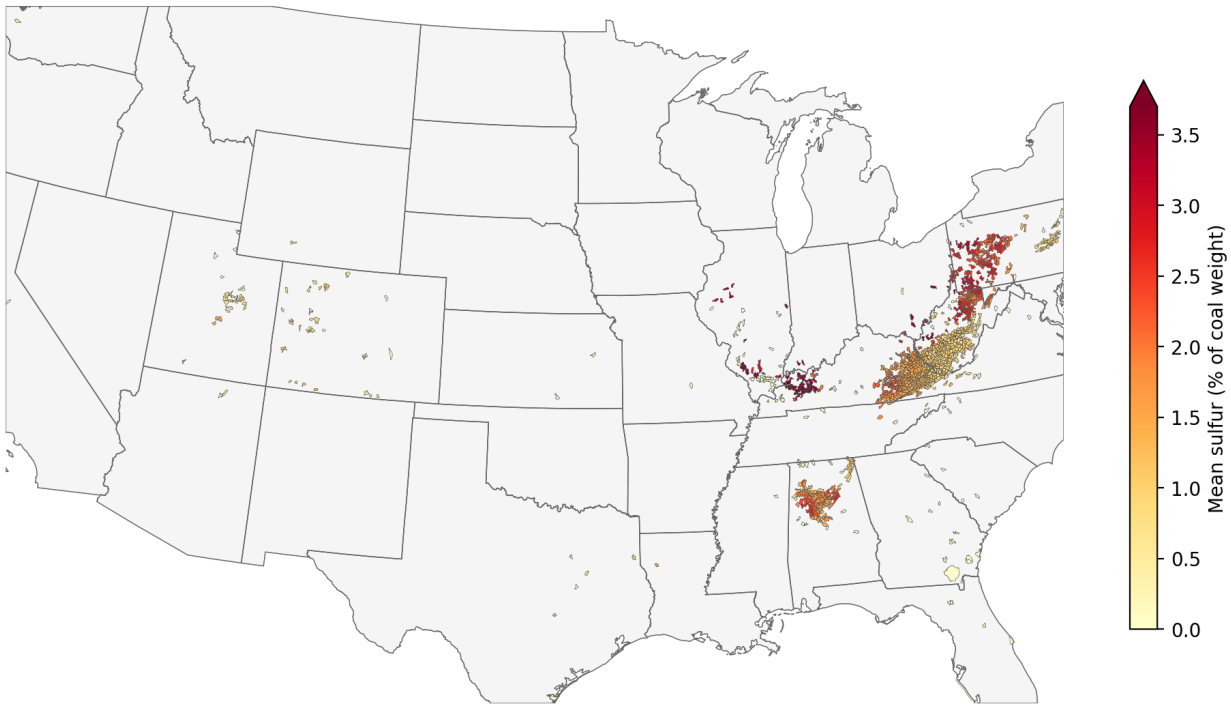


Figure 3: Shading indicates the mean sulfur content of coal, as a percentage of coal weight, for HUC12 watersheds containing an active coal mine. Values are capped at the 95th percentile; watersheds above this threshold appear in the darkest shade, marked by the arrow on the color scale.

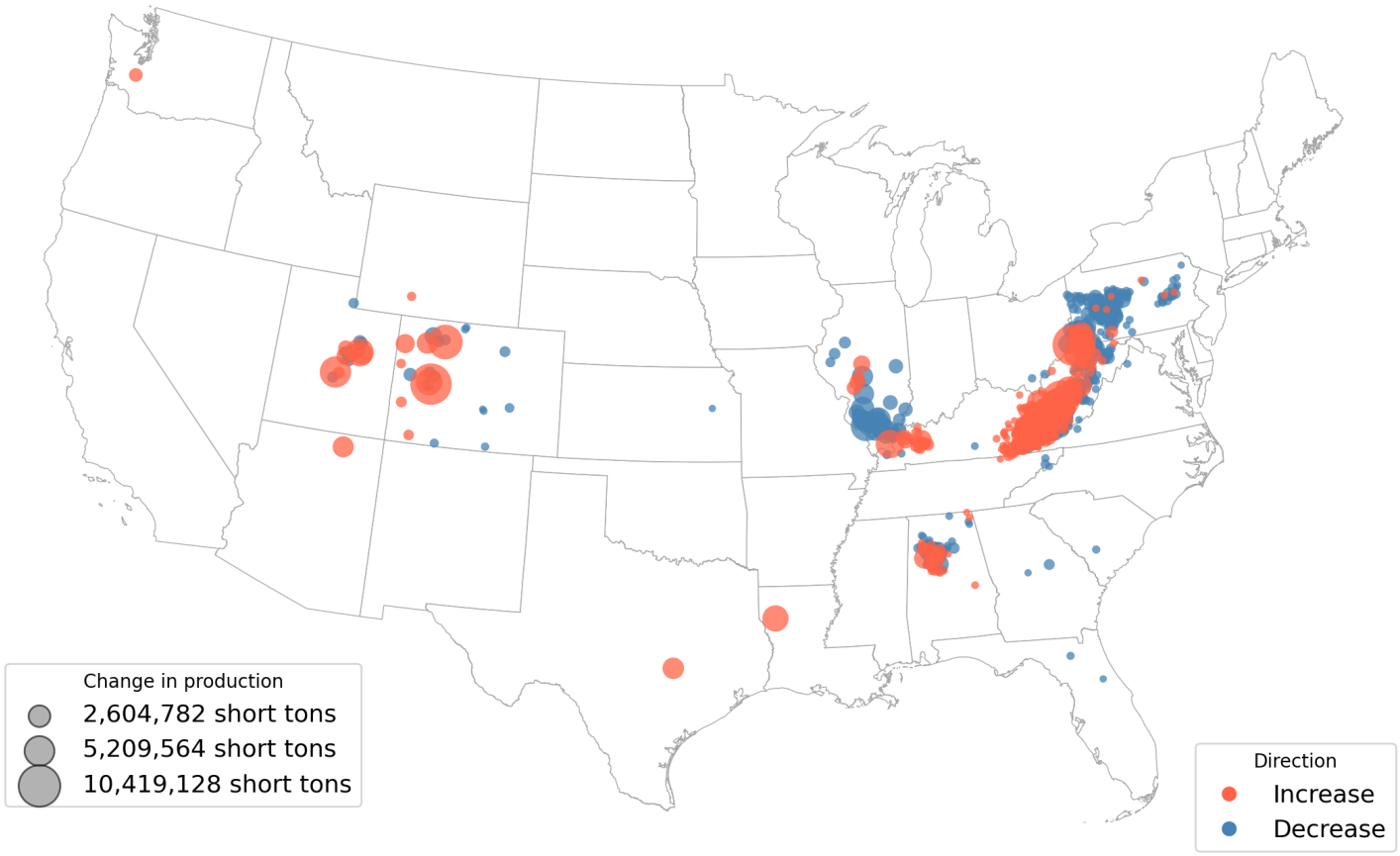


Figure 4: Change in coal production by watershed, 1985 to 2005. Circles are centered on watersheds with at least one active coal mine between 1985 and 2005 and are sized in proportion to the change in total coal production over the period. Red circles indicate an increase in production and blue circles indicate a decrease.

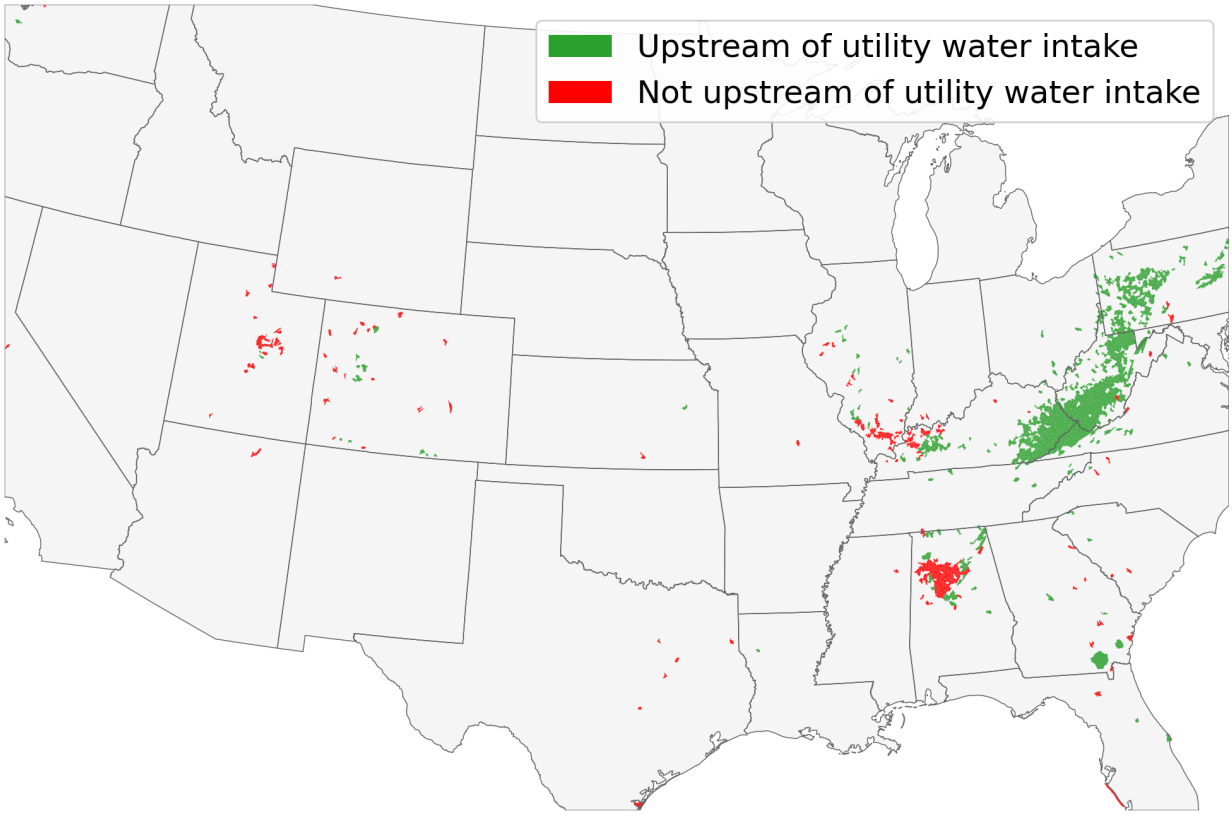


Figure 5: Shading distinguishes HUC12 watersheds that lie directly upstream of a water utility intake and contain a coal mine from HUC12 watersheds containing a coal mine with no water utility located downstream.

Table 5: Balance test of utility characteristics and upstream coal production 1998–2005

	Annual upstream coal prod. (1M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.31		-0.16	
	(0.32)		(0.12)	
Log backcast pop. 1997	0.14	0.19	0.02	0.02
	(0.09)	(0.12)	(0.01)	(0.01)
Log pop. served (SYR2)	-0.12	-0.14	0.01	-0.03
	(0.08)	(0.12)	(0.03)	(0.03)
Number of intake facilities	0.01	-0.04	0.07*	0.07*
	(0.07)	(0.09)	(0.04)	(0.04)
Surface water source	0.15	-0.49	0.12	0.05
	(0.21)	(0.41)	(0.13)	(0.16)
Public ownership	-0.19	-0.07	0.21**	0.10
	(0.32)	(0.28)	(0.09)	(0.11)
Wholesaler	-0.47*	-0.93	-0.22	-0.22
	(0.26)	(1.14)	(0.13)	(0.17)
Source water protection	-0.57	-0.55	-0.28*	-0.15
	(0.34)	(0.54)	(0.15)	(0.19)
State fixed effects		✓		✓
Observations	122	120	122	120
Joint F-test (all covariates)	2.33	0.96	8.96	2.46
p-value	0.03	0.46	$<$0.01	0.02

Notes: Covariates differ in timing, and only population pre-dates the dose window. Backcast population served in 1997 is constructed from decennial census geography and interpolated between the 1990 and 2000 census anchors; it is therefore dated before 1998 by construction but is not an observed 1997 measurement. The remaining covariates (population served, number of intake facilities, number of source HUC12s, primary source type, ownership, wholesaler status, and source water protection) are drawn from the SDWIS/SYR2 inventory and are recorded contemporaneously with the dose window. Heteroskedasticity-robust standard errors. Sample: utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, 1998–2005. *** p$<$0.01, ** p$<$0.05, * p$<$0.1.

Table 6: Effect of annual upstream coal production on inorganic chemicals, SYR2 1998–2005

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Utilities with a coal mine one watershed upstream				
Upstream coal prod. (1M ST)	0.0034***	0.075*	0.0088	0.0018
	(0.0010)	(0.045)	(0.0081)	(0.0012)
Observations	334	432	322	319
Panel B: Utilities with a coal mine within two watersheds upstream				
Upstream coal prod. (1M ST)	0.00038	0.061***	0.014	0.00037
	(0.00028)	(0.015)	(0.012)	(0.00039)
Observations	366	472	353	350

Notes: Within each chemical, the column shows mean measured concentration from the EPA 6-Year Review. Non-detect values replaced by MDL/$\sqrt{2}$ following Ravalli et al. (2022). Explanatory variable is coal production during the year (in millions of short tons) in watersheds upstream of the utility's intake. Panel A: community water systems with a coal mine one watershed upstream and no coal mine colocated with their intake, with production in that upstream watershed. Panel B: community water systems with a coal mine within two watersheds upstream and no coal mine colocated with their intake, with production summed across those watersheds. Standard errors clustered at the utility level. All specifications include utility and HUC02 $\times$ year fixed effects. *** p<0.01, ** p<0.05, * p<0.1.

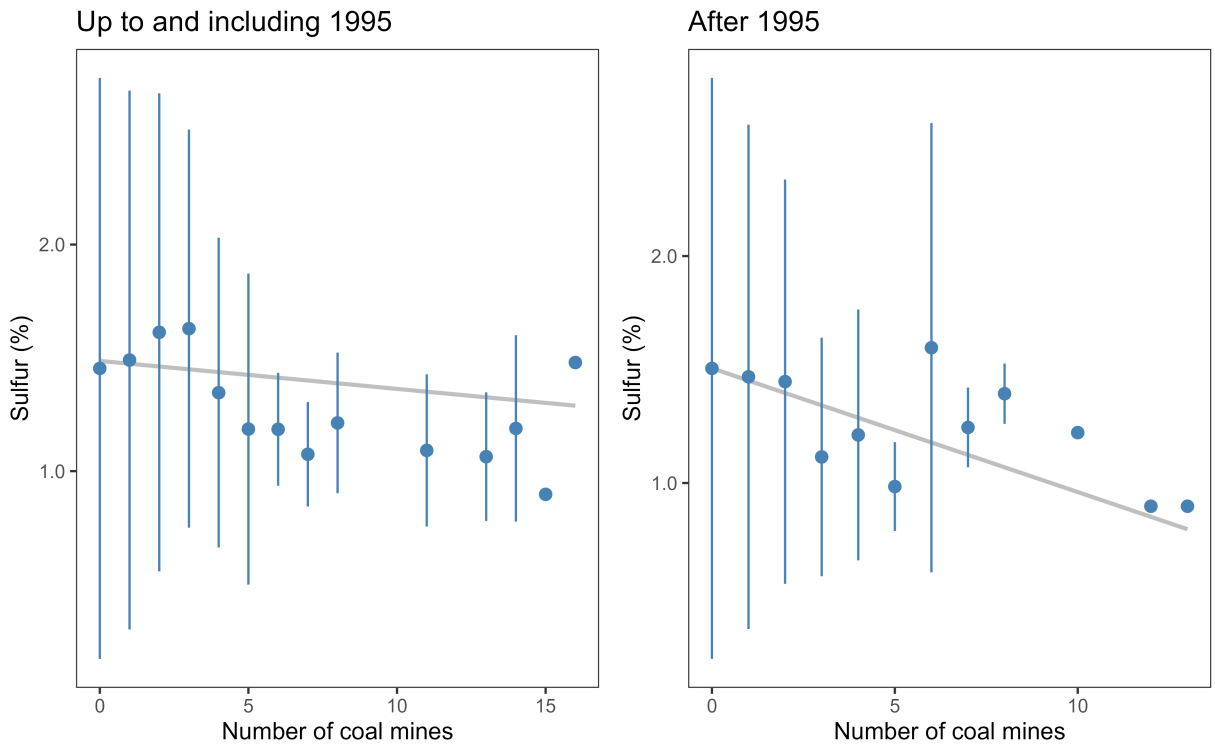


Figure 6: Coal mines and coal sulfur content, before and after 1995. Mean coal sulfur content by number of coal mines in a watershed. Each point is the mean across watershed-year observations with that number of mines; vertical lines show plus or minus one standard deviation; the line is a linear fit of sulfur content on the number of mines, estimated separately for the periods before and after 1995.
Notes: Sample: HUC12 watersheds within two flow steps upstream of a utility water intake, 1985–2005. Points show the mean sulfur as a percent of coal weight within each coal-mine-count bin; ranges show ± 1 standard deviation within bin.

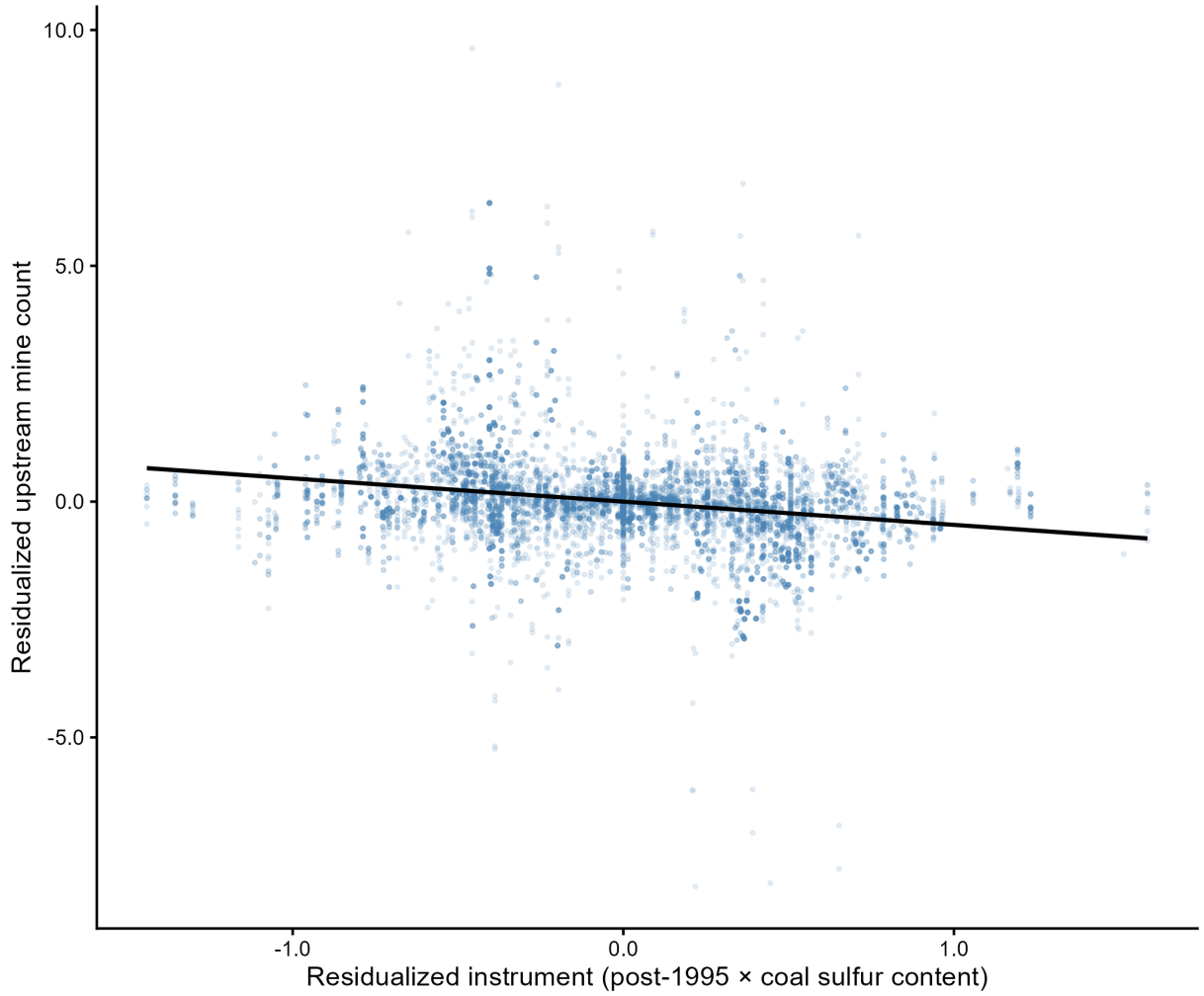


Figure 7: Residualized first-stage relationship. Each point is a utility by year observation for utilities with a coal mine within two watersheds upstream of their intake, 1985 to 2005. Both axes show residuals from separate regressions of each variable on utility, year, and state-by-year fixed effects, retaining only within-utility variation over time. The vertical axis residualizes the number of coal mines upstream within two watersheds; the horizontal axis residualizes the instrument, the post-1995 indicator interacted with the sulfur content of upstream coal. By the Frisch-Waugh-Lovell theorem, the slope of the fitted line equals the first-stage coefficient from the two-stage least squares specification.

Table 7: First stage: effect of the instrument on the number of upstream coal mines, 1985–2005

	Upstream coal mines (sum)
Post-1995 × Upstream sulfur %	-0.49***
	(0.07)
First-stage F-statistic	49.20

Notes: The dependent variable is the number of coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. All specifications include utility and state-by-year fixed effects. Number of observations: 10,641; number of utilities: 565. *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Effect of instrument on utility characteristics

	Number of intake facilities	
	(1)	(2)
Upstream sulfur %		0.99***
		(0.38)
Post-1995 × Upstream sulfur %	0.05	0.06
	(0.04)	(0.04)
Utility fixed effects	✓	
Year fixed effects		✓
State × year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: The dependent variable is the number of active intake facilities operated by the utility in that year. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Sample period 1985–2005. *** p<0.01, ** p<0.05, * p<0.1.

Table 9: Effect of coal mines on inorganic chemical violations at utilities

	Any violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.67**	0.54	0.60*	0.43	0.56*	0.28
	(0.33)	(0.34)	(0.31)	(0.34)	(0.31)	(0.33)
2SLS						
Upstream coal mines (sum)	5.06***	3.89**	4.94***	4.00**	4.56***	3.09*
	(1.70)	(1.75)	(1.57)	(1.56)	(1.64)	(1.67)
RF						
Post-1995 × Upstream sulfur %	-2.09***	-1.92**	-2.04***	-1.97***	-1.89***	-1.53*
	(0.66)	(0.83)	(0.60)	(0.73)	(0.64)	(0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic	38.73	49.20	38.73	49.20	38.73	49.20

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 10: Effect of coal mines on inorganic chemical violations at utilities

	Monitoring and reporting (MR) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.68**	0.55	0.62**	0.42	0.49	0.19
	(0.33)	(0.34)	(0.31)	(0.33)	(0.32)	(0.34)
2SLS						
Upstream coal mines (sum)	5.12***	3.94**	4.85***	3.78**	4.66***	3.06*
	(1.70)	(1.75)	(1.57)	(1.55)	(1.62)	(1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12***	-1.95**	-2.01***	-1.87**	-1.93***	-1.51*
	(0.66)	(0.83)	(0.60)	(0.73)	(0.63)	(0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic	38.73	49.20	38.73	49.20	38.73	49.20

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 11: Effect of coal mines on inorganic chemical violations at utilities

	Maximum contaminant level (MCL) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.01	-0.00	-0.02	0.01	0.05	0.05
	(0.01)	(0.02)	(0.04)	(0.04)	(0.07)	(0.07)
2SLS						
Upstream coal mines (sum)	-0.09	-0.11	0.08*	0.22**	-0.16	-0.20
	(0.07)	(0.17)	(0.04)	(0.10)	(0.28)	(0.26)
RF						
Post-1995 × Upstream sulfur %	0.04	0.05	-0.03**	-0.11**	0.07	0.10
	(0.03)	(0.09)	(0.02)	(0.05)	(0.11)	(0.13)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic	38.73	49.20	38.73	49.20	38.73	49.20

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 12: Effect of coal mining on regulator visit probability by visit type

	Sanitary	Technical assistance	Enforcement	Sample collection	Inspection
	(1)	(2)	(3)	(4)	(5)
OLS					
Upstream coal mines (sum)	0.28	0.10	0.04	-0.07	0.06
	(0.29)	(0.20)	(0.11)	(0.05)	(0.13)
2SLS					
Upstream coal mines (sum)	0.78	0.67	1.07	0.48	2.13**
	(1.20)	(0.95)	(0.81)	(0.50)	(1.07)
RF					
Post-1995 × Upstream sulfur %	-0.39	-0.33	-0.53	-0.24	-1.05**
	(0.59)	(0.47)	(0.39)	(0.24)	(0.49)
First-stage F-statistic	49.20	49.20	49.20	49.20	49.20

Notes: Dependent variable equals 1 if the utility received a regulator visit of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. Sample period 1985–2005. All specifications include utility and state-by-year fixed effects. Number of observations: 10,641; number of utilities: 565. *** p<0.01, ** p<0.05, * p<0.1.

Table 13: Effect of coal mining on enforcement actions by type

	Enforcement					
	Informal		Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.15	-0.06	-1.00***	-0.12	0.33	-0.07
	(0.42)	(0.44)	(0.21)	(0.19)	(0.45)	(0.48)
2SLS						
Upstream coal mines (sum)	-3.06	-0.71	-3.37***	-1.08	2.37	-0.42
	(2.15)	(2.10)	(0.95)	(0.86)	(2.11)	(2.25)
RF						
Post-1995 × Upstream sulfur %	1.26	0.35	1.39***	0.53	-0.98	0.21
	(0.84)	(1.03)	(0.40)	(0.41)	(0.85)	(1.11)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic	38.73	49.20	38.73	49.20	38.73	49.20

Notes: Dependent variable equals 1 if the utility had an enforcement action of that type during the year ("None" equals 1 if it had no enforcement action), 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. Standard errors clustered at the utility level. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 14: Nitrate MR violations following a reading above 50% of the MCL

	Nitrate MR (1-yr)	Nitrate MR (6-mon)
Concen. $>$ 50% MCL	58.93**	26.07**
	(24.55)	(10.51)
Mean concen. (z-score)	1.29	0.08
	(1.46)	(0.91)

Notes: SYR2 sample restricted to utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake (1998–2005), nitrate only. Outcome: nitrate MR (monitoring/reporting) violation in the 365 days following the pollution concentration sample date for the 1-yr column; violation in the 182 days following the pollution concentration sample date for the 6-mon column. Concen. $>$ 50% MCL = reading at 50–100% of the MCL (5 mg/L), which triggers quarterly rather than the baseline annual nitrate testing. Mean concentration = utility-year mean reading, z-scored within chemical. Coefficients and standard errors are in percentage points. All specifications include utility and year fixed effects. Number of observations: 835; number of utilities: 82. Standard errors clustered at the utility level. *** p$<$0.01, ** p$<$0.05, * p$<$0.1.

A Appendix

A.1 Proofs of the propositions

This appendix contains the derivation of the utility's negligence choice and the statements and proofs of the three propositions summarized in Section 2. Notation follows the setup there.

A.1.1 Optimal negligence

The utility's problem is to minimize the losses of the opportunity cost of choosing a negligence below $\bar{a}$ and the expected cost $e(a; c)$. Therefore, the utility maximizes $-\theta[b(\bar{a}) - b(a)] - e(a; c)$ with respect to a. The first-order condition is piecewise:

$$\theta b'(a) = r\, q'(a) \qquad \text{(self-reporting branch)} \tag{6}$$

$$\theta b'(a) = ps\, q'(a) \qquad \text{(under-reporting (MR) branch)} \tag{7}$$

Because $r > ps$, the marginal penalty is lower on the under-reporting branch than on the self-reporting branch.

Which branch a utility takes is not settled by $\hat{c}(a)$ evaluated at either candidate negligence level, because $\hat{c}$ is itself decreasing in the negligence the utility goes on to choose. Proposition 2 shows that the two branch payoffs cross at a single cost draw, and that this crossing point lies between $\hat{c}$ evaluated at the two candidates. Utilities that draw a higher self-reporting cost are more likely to under-report, and a rightward shift in the population's cost distribution raises the share of utilities that do so without moving any type's own crossing point.

A.1.2 The propositions

The model predicts that when water quality deteriorates due to increased pollution, self-reporting rates fall.

Proposition 1: Negligence is increasing in compliance costs. Within each reporting branch,

$$\frac{\partial a}{\partial \theta} = \frac{b'(a)}{-\theta b''(a) + \rho\, q''(a)} > 0,$$

where $\rho \in \{r, ps\}$ is the branch's sanction. Under the interiority condition the first-order condition characterizes the optimum, and the denominator is positive because $b'' < 0$ and $q'' \geq 0$, which is also the second-order condition for the negligence problem within a

branch. Since $r > ps$, a utility that switches from self-reporting to under-reporting faces a lower marginal sanction and moves to a higher negligence level, so negligence rises with θ overall, with an upward jump at the switch.

Proposition 2: The share of under-reporting utilities increases as pollution increases, $\partial MR/\partial m \geq 0$.

For a compliance-cost type θ facing a marginal sanction $\rho > 0$, let $a(\rho;\theta)$ denote the negligence level solving $\theta b'(a) = \rho\, q'(a)$. The negligence a self-reporting utility chooses is $a_{SR}(\theta) \equiv a(r;\theta)$ and the negligence an under-reporting utility chooses is $a_{MR}(\theta) \equiv a(ps;\theta)$, with $a_{SR}(\theta) < a_{MR}(\theta)$ since $r > ps$.

The utility's objective is not globally concave: the marginal penalty drops from $r\,q'(a)$ to $ps\,q'(a)$ at the negligence level where $\hat{c}(a) = c$. Writing the utility's problem as a maximization, $\max_a\{\theta b(a) - \theta b(\bar{a}) - \min[\, c + rq(a),\ \ t + ps\,q(a)\,]\}$, the inner minimum enters with a negative sign and is therefore a maximum over the two reporting branches, which can be exchanged with the maximization over a: $\max_a \max\{\cdot,\cdot\} = \max\{\max_a(\cdot), \max_a(\cdot)\}$, since neither branch's maximizer depends on the other's value. So the value of the problem is the larger of two per-branch maxima, each built from $W(\rho;\theta) \equiv \max_{a\in[0,\bar{a}]}\{\theta b(a) - \rho\, q(a)\}$ for $\rho \in \{r, ps\}$, each attained at a unique $a(\rho;\theta)$ since $\theta b(a) - \rho\, q(a)$ is strictly concave. No negligence level besides $a_{SR}(\theta) \equiv a(r;\theta)$ and $a_{MR}(\theta) \equiv a(ps;\theta)$ is ever optimal, for any c. The payoff from self-reporting is $W(r;\theta) - \theta b(\bar{a}) - c$ and the payoff from under-reporting is $W(ps;\theta) - \theta b(\bar{a}) - t$, and the utility takes whichever is larger. The former falls one-for-one in c and the latter does not involve c at all, so their difference is affine in c with nonzero slope and crosses zero exactly once. Cancelling $\theta b(\bar{a})$ and applying the envelope theorem, $\partial W(\rho;\theta)/\partial\rho = -q(a(\rho;\theta))$, the crossing point is the type-specific switching cost

$$c^*(\theta) \equiv t + W(r;\theta) - W(ps;\theta) = t - \int_{ps}^{r} q\big(a(\rho;\theta)\big)\, d\rho. \tag{8}$$

Since $q(a(\rho;\theta)) \leq q(\bar{a})$ for every ρ, the maintained assumption $t \geq (r - ps)q(\bar{a})$ delivers $c^*(\theta) \geq 0$. When negligence carries no risk of a water quality violation, $q \equiv 0$, the integral vanishes and the cutoff is simply the under-reporting penalty t. The cutoff depends only on the sanctions and on the utility's own compliance-cost type, not on pollution m.

A type-θ utility under-reports if and only if its cost draw exceeds this cutoff, $c > c^*(\theta)$, so the share of utilities under-reporting and receiving an MR violation is

$$MR(m) = \int_0^{\bar{\theta}} \big[\, 1 - G(c^*(\theta);m)\,\big]\, f(\theta;m)\, d\theta \ \equiv\ \int_0^{\bar{\theta}} h(\theta,m)\, f(\theta;m)\, d\theta. \tag{9}$$

Differentiating (9) splits into two channels:

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m)\, d\theta}_{\text{Channel A: cost-distribution shift}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m}\, d\theta}_{\text{Channel B: type-distribution shift}}. \tag{10}$$

Channel A:

$$\frac{\partial h(\theta, m)}{\partial m} = -\frac{\partial G(c^*(\theta); m)}{\partial m} \geq 0 \quad \text{for every } \theta,$$

since $\partial G(\cdot\,; m)/\partial m \leq 0$ pointwise.

Channel A is therefore non-negative: holding the type composition fixed, a rightward shift in the self-reporting cost distribution pushes more mass of c above each type's unchanged cutoff.

Channel B:

The maintained assumption signs $\partial F(\theta; m)/\partial m \leq 0$, but Channel B is written in terms of the *density* derivative $\partial f(\theta; m)/\partial m$, which has no assumed sign — a rightward shift in a cdf can raise the density in one part of the support and lower it in another. To use the assumption we actually have, rewrite Channel B in terms of F instead of f via integration by parts.

Define $\varphi(\theta; m) \equiv \partial F(\theta; m)/\partial m$. Since $f(\theta; m) = \partial F(\theta; m)/\partial \theta$, swapping the order of the two partial derivatives gives $\partial f(\theta; m)/\partial m = \partial \varphi(\theta; m)/\partial \theta$, so Channel B is

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial \varphi(\theta; m)}{\partial \theta}\, d\theta.$$

Integrating by parts (treating $[\partial \varphi(\theta; m)/\partial \theta]\, d\theta$ as $d\varphi$),

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial \varphi(\theta; m)}{\partial \theta}\, d\theta = \Big[\, h(\theta, m)\, \varphi(\theta; m) \,\Big]_0^{\bar{\theta}} - \int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial \theta}\, \varphi(\theta; m)\, d\theta.$$

The boundary term vanishes: $\varphi(0; m) = \partial F(0; m)/\partial m = 0$ and $\varphi(\bar{\theta}; m) = \partial F(\bar{\theta}; m)/\partial m = 0$, since $F(0; m) = 0$ and $F(\bar{\theta}; m) = 1$ for every m (the cdf is pinned at the endpoints of its support regardless of m). Substituting $\varphi(\theta; m) = \partial F(\theta; m)/\partial m$ back in,

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m}\, d\theta = -\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial \theta} \frac{\partial F(\theta; m)}{\partial m}\, d\theta. \tag{11}$$

Signing (11) therefore requires the slope of h in θ. Differentiating (8) and applying the envelope theorem in θ, $\partial W(\rho; \theta)/\partial \theta = b(a(\rho; \theta))$,

$$c^{*\prime}(\theta) = b(a_{SR}(\theta)) - b(a_{MR}(\theta)) < 0,$$

since $a_{SR}(\theta) < a_{MR}(\theta)$ and $b' > 0$. Then

$$\frac{\partial h(\theta, m)}{\partial \theta} = -g(c^*(\theta); m)\, c^{*\prime}(\theta) \geq 0,$$

so h is non-decreasing in θ: higher-cost types choose more negligence, which lowers their own switching cutoff and makes them more likely to under-report.

$$\text{Channel B} = -\int_0^{\bar{\theta}} \underbrace{\frac{\partial h(\theta, m)}{\partial \theta}}_{\geq 0} \underbrace{\frac{\partial F(\theta; m)}{\partial m}}_{\leq 0} d\theta \;\geq\; 0.$$

Both channels are therefore non-negative, so $\partial MR/\partial m = \text{Channel A} + \text{Channel B} \geq 0$.

Pollution from mining can raise MR violations by increasing self-reporting costs relative to each type's fixed cutoff and by shifting the population toward higher-cost types who are already structurally closer to their own cutoff.

Proposition 3: Increasing all penalties increases the share of utilities self-reporting.

Consider scaling every sanction by a common factor $\lambda > 0$: $(r, s, t) \to \lambda(r, s, t)$, holding $q(\cdot)$, $b(\cdot)$, p, and the cost draws θ, c fixed. Costs become

$$\begin{aligned} \text{self-report:} \quad & c + q(a)\, \lambda r \\ \text{under-report (MR):} \quad & \lambda t + p\, [q(a)\, \lambda s] \end{aligned}$$

Differentiating the expected annual cost in a on each branch,

$$\begin{aligned} \frac{\partial e_\lambda(a; c)}{\partial a} &= \lambda r\, q'(a) = \lambda\, \frac{\partial e(a; c)}{\partial a} && \text{(self-reporting branch)} \\ \frac{\partial e_\lambda(a; c)}{\partial a} &= \lambda p s\, q'(a) = \lambda\, \frac{\partial e(a; c)}{\partial a} && \text{(under-reporting (MR) branch)} \end{aligned}$$

Negligence falls in λ.

$$\frac{\partial a}{\partial \lambda} = \frac{\rho\, q'(a)}{\theta b''(a) - \lambda \rho\, q''(a)}.$$

The second-order condition invoked in Proposition 1, $\lambda \rho\, q''(a) - \theta b''(a) \geq 0$, makes the denominator non-positive, so $\partial a/\partial \lambda \leq 0$: negligence is weakly decreasing in the penalty scale, within each branch.

The reporting rule scales by λ. The self-report condition with the scaled penalties, $c + \lambda r\, q(a) \leq \lambda[t + ps\, q(a)]$ rearranges to

$$c \leq \hat{c}_\lambda(a) \equiv \lambda[\, t - (r - ps)\, q(a)\,] = \lambda\, \hat{c}(a). \tag{12}$$

Unlike $\hat{c}_\lambda(a)$ itself, the self-report cost c does not scale with λ since it is a technology/labor cost, not a legal penalty, so scaling λ up moves the threshold $\hat{c}_\lambda(a)$ relative to a fixed distribution of c.

Scaling the sanctions replaces the marginal sanction pair (r, ps) with $(\lambda r, \lambda ps)$ and the under-reporting penalty t with λt, so the switching cost of (8) becomes

$$C^*(\lambda;\theta) = \lambda t - \int_{\lambda ps}^{\lambda r} q(a(u;\theta))\, du,$$

with derivative

$$\frac{\partial C^*(\lambda;\theta)}{\partial \lambda} = t - \Big[\, r\, q(a(\lambda r;\theta)) - ps\, q(a(\lambda ps;\theta))\Big] \;\geq\; t - (r - ps)\, q(\bar{a}) \;\geq\; 0.$$

The first inequality uses $a(\lambda r;\theta) \leq a(\lambda ps;\theta)$, so that $q(a(\lambda r;\theta)) \leq q(a(\lambda ps;\theta))$ and the bracketed term is at most $(r - ps)\, q(a(\lambda r;\theta)) \leq (r - ps)\, q(\bar{a})$. The second is the maintained assumption on t. Raising the penalty scale therefore raises the switching cost of every type, holding the population's cost draws fixed. Aggregating over types, the share of utilities that self-report,

$$SR(\lambda) = \int_0^{\bar{\theta}} G(C^*(\lambda;\theta); m)\, f(\theta; m)\, d\theta,$$

is non-decreasing in λ, since G is increasing in its first argument and $\partial C^*(\lambda;\theta)/\partial\lambda \geq 0$ for every θ. A higher penalty scale reduces negligence and increases self-reporting.

A.2 Robustness tests

A.2.1 Coal mines two and three watersheds upstream

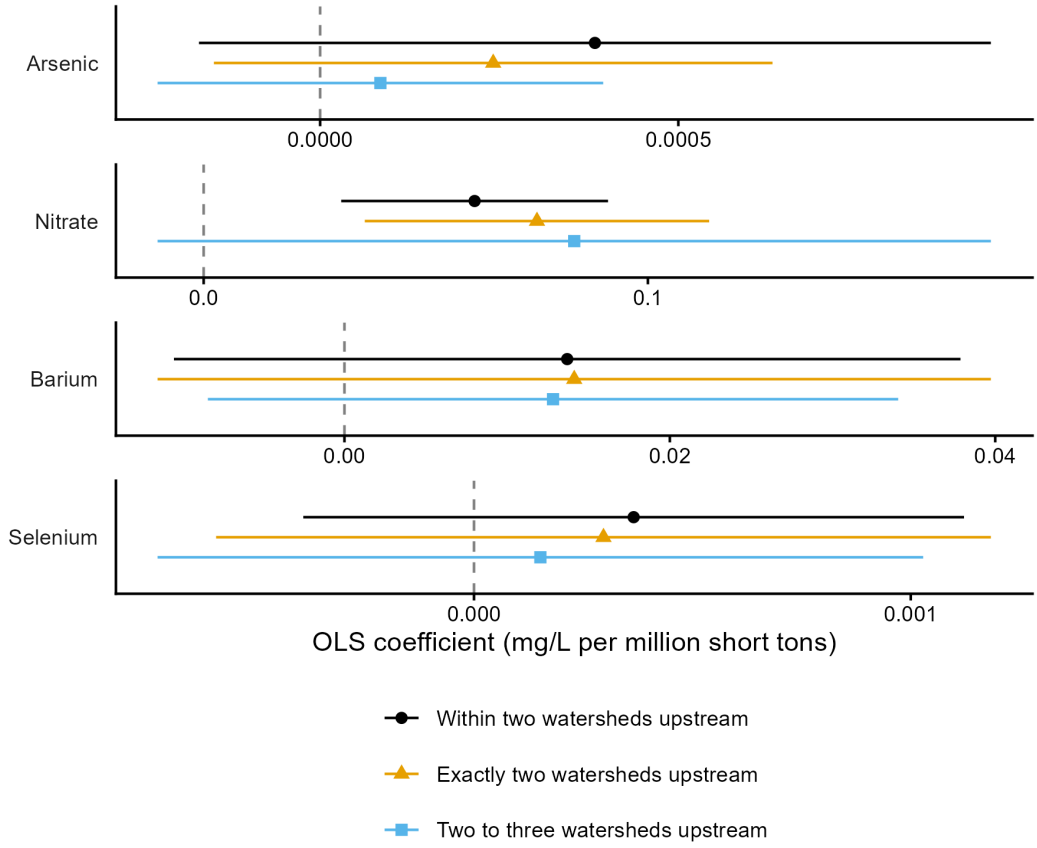


Figure A1: Effect of coal production on measured pollution concentrations, by distance of the mines upstream

Notes: Estimates use utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it. Black: coal production within two watersheds upstream. Orange: production in watersheds exactly two watersheds upstream. Blue: production in watersheds two or three watersheds upstream. Each estimate is ordinary least squares of the mean measured concentration from the Six-Year Review (1998–2005) on annual coal production in millions of short tons in those watersheds, with utility and region-by-year fixed effects. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.

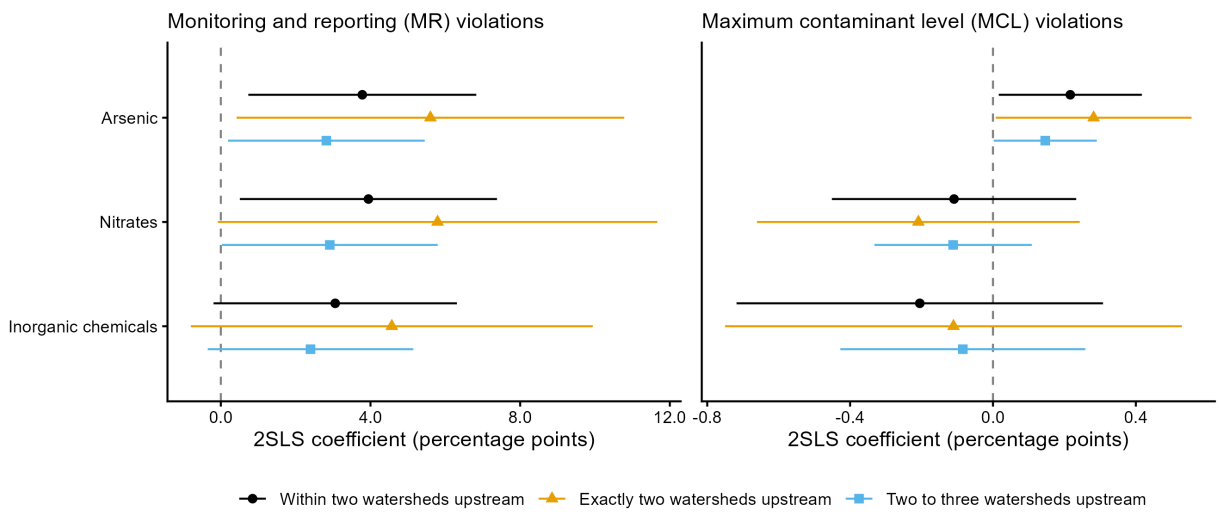


Figure A2: Effect of coal mines on inorganic chemical violations, by distance of the mines upstream

Notes: All three estimates use the same sample, utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it, 1985–2005. Black: active coal mines within two watersheds upstream. Orange: active coal mines in watersheds exactly two watersheds upstream. Blue: active coal mines in watersheds two or three watersheds upstream. Utilities with no mine in those watersheds have zero. Each estimate is two-stage least squares, instrumenting with the post-1995 indicator interacted with the average coal sulfur content of the same set of watersheds, with utility and state-by-year fixed effects. The dependent variable equals 1 if the violation occurred during the year, and coefficients are in percentage points. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.

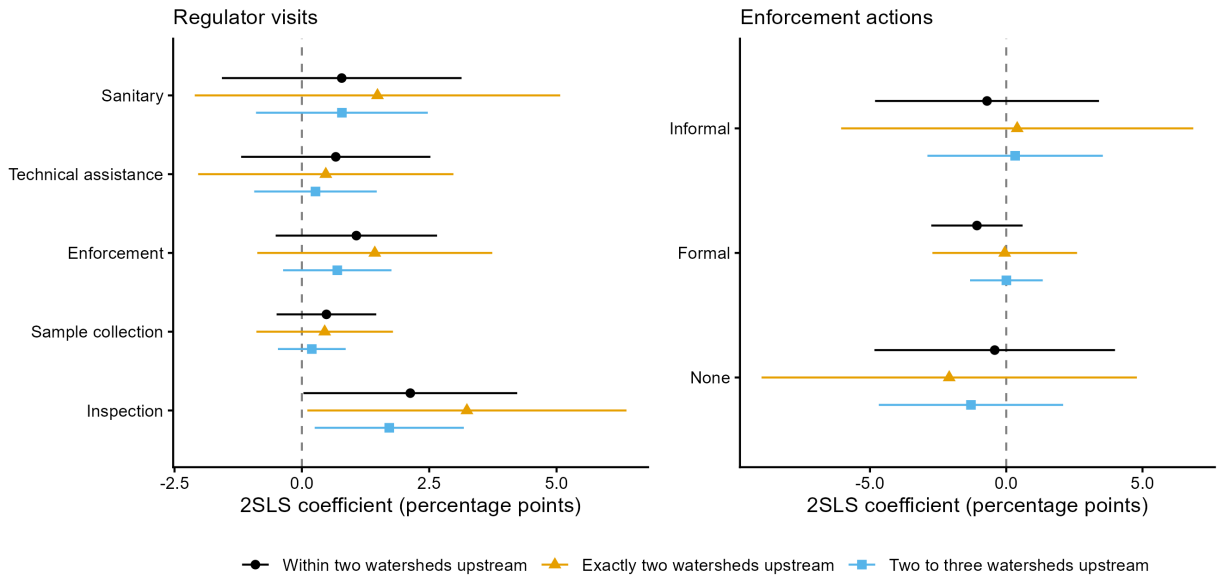


Figure A3: Effect of coal mines on regulator visits and enforcement actions, by distance of the mines upstream

Notes: All three estimates use the same sample, utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it, 1985–2005. Black: active coal mines within two watersheds upstream. Orange: active coal mines in watersheds exactly two watersheds upstream. Blue: active coal mines in watersheds two or three watersheds upstream. Utilities with no mine in those watersheds have zero. Each estimate is two-stage least squares, instrumenting with the post-1995 indicator interacted with the average coal sulfur content of the same set of watersheds, with utility and state-by-year fixed effects. The dependent variable equals 1 if the visit or enforcement action occurred during the year, and coefficients are in percentage points. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.

A.2.2 Over-identified IV: upstream and downstream sulfur instruments

Table 15: First stage: effect of the instruments on the number of upstream coal mines, 1985–2005

	Upstream coal mines (sum)		
	(1)	(2)	(3)
Post-1995 × Upstream sulfur %	-0.49***		-0.47***
	(0.07)		(0.09)
Post-1995 × Downstream sulfur %		-0.34***	-0.03
		(0.06)	(0.06)
First-stage F-statistic	49.20	37.28	24.68

Notes: The dependent variable is the number of coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. The first-stage F-statistic is clustered at the utility level; in column 3 it tests both instruments jointly. Standard errors clustered at the utility level. Sample period 1985–2005. All specifications include utility and state-by-year fixed effects. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 16: Effect of instruments on utility characteristics

	Number of intake facilities	
	(1)	(2)
Upstream sulfur %		-1.18
		(2.70)
Downstream sulfur %		2.34
		(3.09)
Post-1995 × Upstream sulfur %	0.07	0.06
	(0.06)	(0.05)
Post-1995 × Downstream sulfur %	-0.02	0.00
	(0.04)	(0.04)
Utility fixed effects	✓	
Year fixed effects		✓
State × year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: The dependent variable is the number of active intake facilities operated by the utility in that year. Upstream and downstream sulfur are the average coal sulfur content of watersheds within two flow steps upstream and downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Sample period 1985–2005. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 17: Effect of upstream coal mines on inorganic chemical violations at utilities, two instruments

	Any violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.67**	0.54	0.60*	0.43	0.56*	0.28
	(0.33)	(0.34)	(0.31)	(0.34)	(0.31)	(0.33)
2SLS						
Upstream coal mines (sum)	5.05***	3.90**	4.89***	4.04***	4.52***	3.13*
	(1.70)	(1.74)	(1.56)	(1.56)	(1.63)	(1.66)
RF						
Post-1995 × Upstream sulfur %	-2.23**	-1.66	-1.75*	-1.29	-1.60*	-0.89
	(1.11)	(1.30)	(0.92)	(1.05)	(0.91)	(1.05)
Post-1995 × Downstream sulfur %	0.15	-0.28	-0.31	-0.75	-0.31	-0.70
	(0.89)	(0.93)	(0.74)	(0.78)	(0.72)	(0.74)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic (joint)	19.95	24.68	19.95	24.68	19.95	24.68
Overidentification test p-value	0.95	0.86	0.49	0.41	0.49	0.40

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Upstream coal mines are instrumented with two instruments. The first-stage F-statistic (clustered at the utility level) tests the two instruments jointly. The overidentification test is Hansen's J statistic, robust to clustering at the utility level, for the null hypothesis that both instruments are valid. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 18: Effect of upstream coal mines on inorganic chemical violations at utilities, two instruments

	Monitoring and reporting (MR) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.68**	0.55	0.62**	0.42	0.49	0.19
	(0.33)	(0.34)	(0.31)	(0.33)	(0.32)	(0.34)
2SLS						
Upstream coal mines (sum)	5.11***	3.96**	4.82***	3.82**	4.60***	3.11*
	(1.71)	(1.74)	(1.56)	(1.55)	(1.62)	(1.66)
RF						
Post-1995 × Upstream sulfur %	-2.20**	-1.63	-1.78*	-1.25	-1.47	-0.67
	(1.11)	(1.30)	(0.92)	(1.05)	(0.90)	(1.04)
Post-1995 × Downstream sulfur %	0.08	-0.34	-0.24	-0.68	-0.49	-0.92
	(0.88)	(0.93)	(0.74)	(0.77)	(0.70)	(0.73)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic (joint)	19.95	24.68	19.95	24.68	19.95	24.68
Overidentification test p-value	0.89	0.81	0.56	0.45	0.33	0.24

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Upstream coal mines are instrumented with two instruments. The first-stage F-statistic (clustered at the utility level) tests the two instruments jointly. The overidentification test is Hansen's J statistic, robust to clustering at the utility level, for the null hypothesis that both instruments are valid. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 19: Effect of upstream coal mines on inorganic chemical violations at utilities, two instruments

	Maximum contaminant level (MCL) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.01	-0.00	-0.02	0.01	0.05	0.05
	(0.01)	(0.02)	(0.04)	(0.04)	(0.07)	(0.07)
2SLS						
Upstream coal mines (sum)	-0.08	-0.11	0.08*	0.22**	-0.14	-0.22
	(0.06)	(0.18)	(0.04)	(0.10)	(0.27)	(0.27)
RF						
Post-1995 × Upstream sulfur %	-0.02	0.01	0.03	-0.04	-0.10	-0.09
	(0.07)	(0.07)	(0.04)	(0.05)	(0.15)	(0.13)
Post-1995 × Downstream sulfur %	0.06	0.04	-0.07	-0.07	0.18	0.20
	(0.09)	(0.09)	(0.05)	(0.05)	(0.16)	(0.18)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic (joint)	19.95	24.68	19.95	24.68	19.95	24.68
Overidentification test p-value	0.53	0.64	0.12	0.20	0.27	0.23

Notes: Dependent variable equals 1 if the utility had a violation of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Upstream coal mines are instrumented with two instruments. The first-stage F-statistic (clustered at the utility level) tests the two instruments jointly. The overidentification test is Hansen's J statistic, robust to clustering at the utility level, for the null hypothesis that both instruments are valid. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 20: Effect of upstream coal mining on regulator visit probability by visit type, two instruments

	Sanitary	Technical assistance	Enforcement	Sample collection	Inspection
	(1)	(2)	(3)	(4)	(5)
OLS					
Upstream coal mines (sum)	0.28	0.10	0.04	-0.07	0.06
	(0.29)	(0.20)	(0.11)	(0.05)	(0.13)
2SLS					
Upstream coal mines (sum)	0.82	0.66	1.09	0.50	2.09*
	(1.20)	(0.95)	(0.80)	(0.49)	(1.07)
RF					
Post-1995 × Upstream sulfur %	0.19	-0.34	-0.24	0.07	-1.55**
	(0.93)	(0.61)	(0.57)	(0.43)	(0.65)
Post-1995 × Downstream sulfur %	-0.64	0.02	-0.31	-0.34	0.55
	(0.80)	(0.51)	(0.38)	(0.34)	(0.51)
First-stage F-statistic (joint)	24.68	24.68	24.68	24.68	24.68
Overidentification test p-value	0.46	0.94	0.47	0.34	0.22

Notes: Dependent variable equals 1 if the utility received a regulator visit of that type during the year, 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Upstream coal mines are instrumented with two instruments. The first-stage F-statistic (clustered at the utility level) tests the two instruments jointly. The overidentification test is Hansen's J statistic, robust to clustering at the utility level, for the null hypothesis that both instruments are valid. Sample period 1985–2005. All specifications include utility and state-by-year fixed effects. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 21: Effect of upstream coal mining on enforcement actions by type, two instruments

	Enforcement					
	Informal		Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.15	-0.06	-1.00***	-0.12	0.33	-0.07
	(0.42)	(0.44)	(0.21)	(0.19)	(0.45)	(0.48)
2SLS						
Upstream coal mines (sum)	-2.94	-0.79	-3.36***	-1.10	2.31	-0.36
	(2.14)	(2.09)	(0.95)	(0.85)	(2.12)	(2.24)
RF						
Post-1995 × Upstream sulfur %	0.12	-0.88	1.41**	0.18	-0.42	1.00
	(1.41)	(1.53)	(0.61)	(0.57)	(1.58)	(1.68)
Post-1995 × Downstream sulfur %	1.23	1.34	-0.01	0.39	-0.61	-0.86
	(1.19)	(1.15)	(0.53)	(0.44)	(1.35)	(1.27)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
First-stage F-statistic (joint)	19.95	24.68	19.95	24.68	19.95	24.68
Overidentification test p-value	0.25	0.24	0.81	0.42	0.59	0.48

Notes: Dependent variable equals 1 if the utility had an enforcement action of that type during the year ("None" equals 1 if it had no enforcement action), 0 otherwise; coefficients and standard errors are multiplied by 100 to show percentage point change. Upstream coal mines is the number of active coal mines in watersheds within two flow steps upstream of the utility's intake, summed across those watersheds. The two instruments interact an indicator for the post-1995 period with the average coal sulfur content of watersheds within two flow steps upstream and, separately, within two flow steps downstream of the utility's intake. The sample is utilities with a coal mine within two flow steps upstream of their intake and no coal mine colocated with their intake, in states with at least one other such utility. Standard errors clustered at the utility level. Upstream coal mines are instrumented with two instruments. The first-stage F-statistic (clustered at the utility level) tests the two instruments jointly. The overidentification test is Hansen's J statistic, robust to clustering at the utility level, for the null hypothesis that both instruments are valid. Sample period 1985–2005. Number of observations: 10,641; number of utilities: 565. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

A.2.3 Upstream coal mines versus upstream coal production

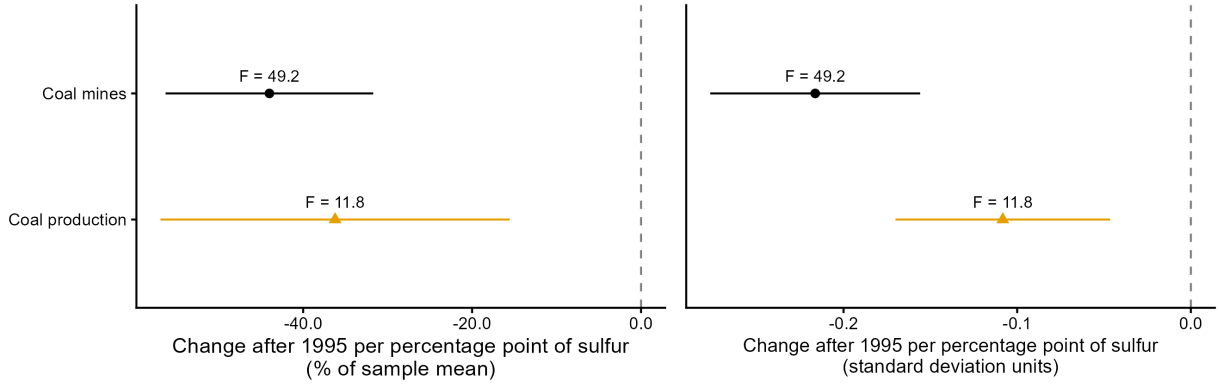


Figure A4: Effect of the post-1995 sulfur instrument on upstream coal mines and upstream coal production

Notes: The sample is utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it, in states with at least one other such utility, 1985–2005. Coal mines and coal production are the number of active mines and annual coal production in millions of short tons, summed over watersheds within two watersheds upstream. The explanatory variable is the post-1995 indicator interacted with the average coal sulfur content of the same watersheds, with utility and state-by-year fixed effects. The left panel divides the coefficient by the sample mean of the dependent variable, and the right panel divides it by the dependent variable's standard deviation. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.

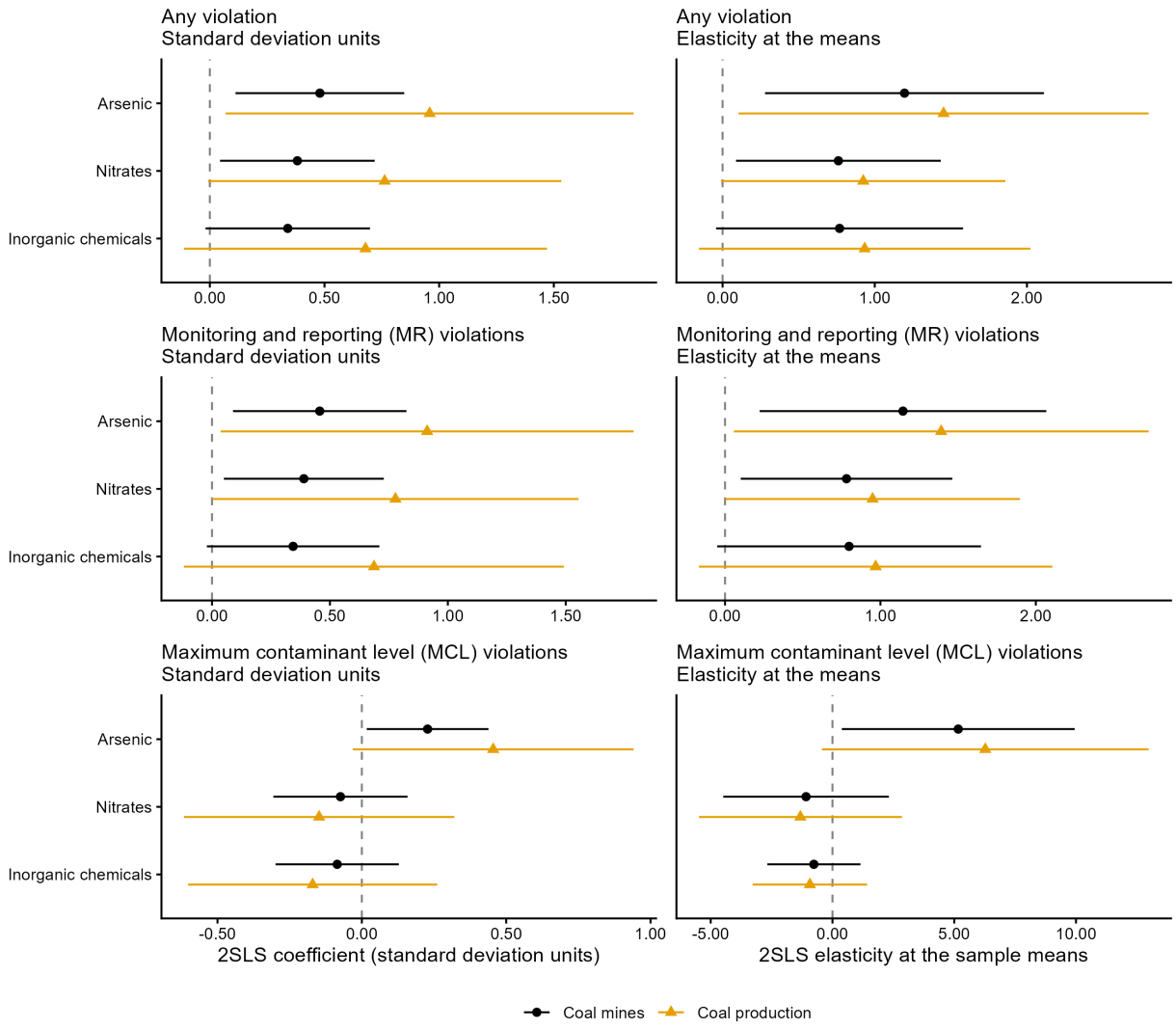


Figure A5: Effect of upstream coal mines and upstream coal production on inorganic chemical violations

Notes: The sample is utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it, in states with at least one other such utility, 1985–2005. Each estimate is two-stage least squares, instrumenting the regressor, either the number of active coal mines or annual coal production, summed over watersheds within two watersheds upstream, with the post-1995 indicator interacted with the average coal sulfur content of the same watersheds, with utility and state-by-year fixed effects. The dependent variable equals 1 if the violation occurred during the year. In the left panels the dependent variable and the regressor are each in standard deviation units. In the right panels each is divided by its sample mean, so coefficients are elasticities evaluated at the sample means. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.

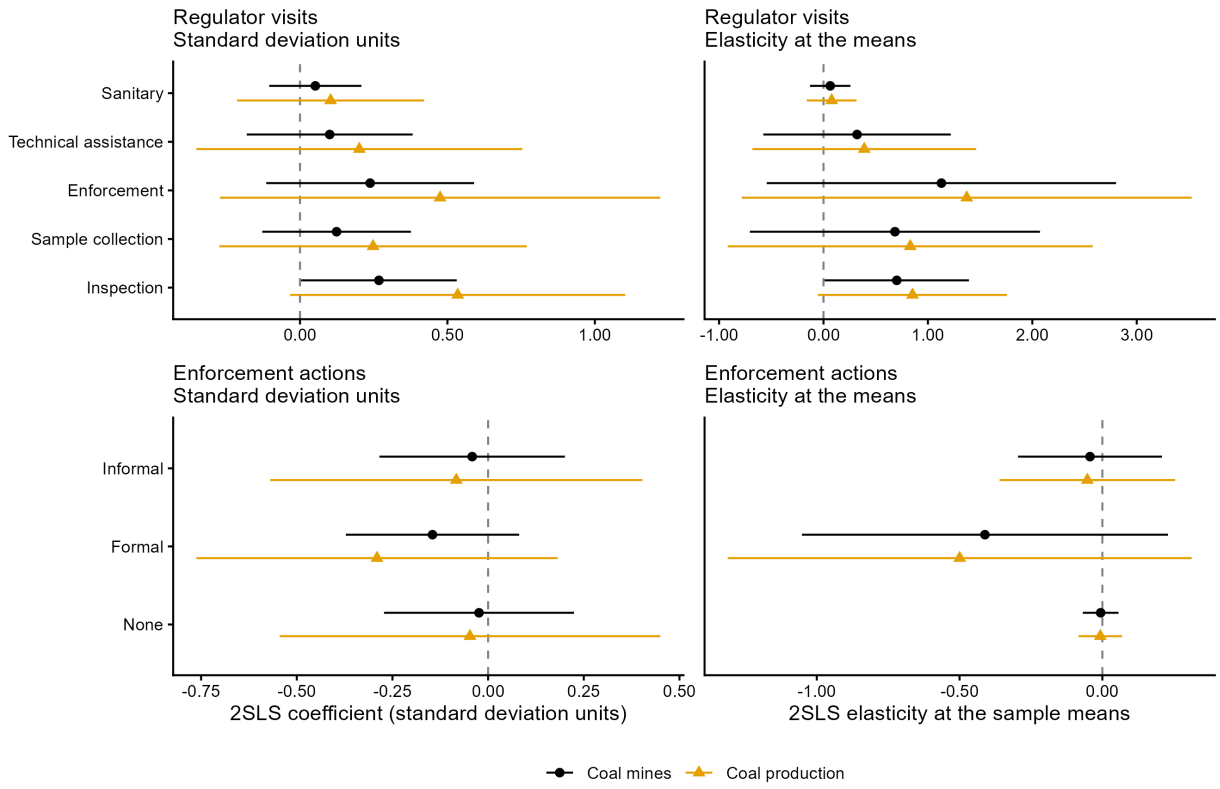


Figure A6: Effect of upstream coal mines and upstream coal production on regulator visits and enforcement actions

Notes: The sample is utilities with a coal mine within two watersheds upstream of their intake and no coal mine colocated with it, in states with at least one other such utility, 1985–2005. Each estimate is two-stage least squares, instrumenting the regressor, either the number of active coal mines or annual coal production, summed over watersheds within two watersheds upstream, with the post-1995 indicator interacted with the average coal sulfur content of the same watersheds, with utility and state-by-year fixed effects. The dependent variable equals 1 if the visit or enforcement action occurred during the year, and for the no-enforcement outcome it equals 1 if the utility had no enforcement action during the year. In the left panels the dependent variable and the regressor are each in standard deviation units. In the right panels each is divided by its sample mean, so coefficients are elasticities evaluated at the sample means. Bars are 95 percent confidence intervals with standard errors clustered at the utility level.